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Solar power energy system with rankine engine

Abstract

The systems of the present disclosure include a solar-powered steam Rankine cycle (SRC) subsystem to convert solar energy into thermal energy and store the thermal energy; an ejector refrigeration cycle (ERC) subsystem to provide a first refrigeration effect with a first range of temperature based on the thermal energy; an absorption refrigeration cycle (ARC) subsystem to provide a second refrigeration effect with a second range of temperature based on the thermal energy; a brine refrigeration cycle (BRC) subsystem to generate and store when there is no cooling demand and provide a third refrigeration effect with a third range of temperature based on the electrical power generated by the ERC subsystem and the ice being melted; and an adsorption refrigeration cycle (ADRC) subsystem to provide a fourth refrigeration effect with a fourth range of temperature based on the thermal energy.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation of U.S. application Ser. No. 18/083,685, now allowed, having a filing date of Dec. 19, 2022.

TECHNICAL FIELD
(1) The present disclosure relates to refrigeration systems and more particularly relates to solar energy powered refrigeration combined with an ice storage system.
DISCUSSION OF RELATED ART
(2) The "background" description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description which may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present invention.
(3) Refrigeration systems are the major consumer of electricity. Use of refrigeration and air conditioners in residential and commercial sectors has increased significantly in recent years. Globally residential and commercial sectors account for about 40% of total energy use. Both power generator and vapor compression refrigeration systems operate at lower efficiencies during peak time (high temperature and humidity), thereby resulting in increased peak load on an electric grid. The vapor compression refrigeration systems are predominantly used for air cooling purposes which is energy-intensive in a hot and humid country.
(4) Solar cooling is known as an efficient technology that provides environmental benefits including reducing main grid demand and shifting the load during peak usage while reducing greenhouse gas emissions. Solar driven refrigeration and air conditioning systems are a favorable technology due to the need for high refrigeration demand during times (daylight) when the availability of solar energy is maximum. Some applications require a high levels of cooling, and a single refrigeration unit is not known in the art to address such cooling levels.
(5) Accordingly, it is one object of the present disclosure to provide systems and methods for solar energy powered refrigeration combined with an ice storage system which is operated in three cycles, that is, a solar energy cycle, a refrigeration cycle, and a cooling demand application to achieve a required temperature.

SUMMARY

(6) According to one aspect of the present disclosure, a system is described which includes a solar-powered steam Rankine cycle (SRC) subsystem, an ejector refrigeration cycle (ERC) subsystem, an absorption refrigeration cycle (ARC) subsystem, a brine refrigeration cycle (BRC) subsystem, and an adsorption refrigeration cycle (ADRC) subsystem. The solar-powered SRC subsystem is configured to receive solar energy, convert the solar energy into thermal energy, and store the thermal energy. The ERC subsystem is configured to generate an electrical power based on the thermal energy and provide a first refrigeration effect with a first range of temperature based on the thermal energy. The ARC subsystem is configured to provide a second refrigeration effect with a second range of temperature based on the thermal energy. The BRC subsystem is configured to generate ice when there is no cooling demand and provide a third refrigeration effect with a third range of temperature based on the electrical power generated by the ERC subsystem and the ice being melted. The ADRC subsystem is configured to provide a fourth refrigeration effect with a fourth range of temperature based on the thermal energy.

(7) In an aspect of the present disclosure, the SRC subsystem includes a central receiver (CR), a heat recovery vapor generator (HRVG), a hot molten salt storage tank, a cold molten salt storage tank, and a heat transfer fluid (HTF) pump. The CR is configured to receive the solar energy to heat cold molten salt flowing through the central receiver. The HRVG is configured to generate superheated steam for the ERC subsystem based on the heated molten salt and pass the heated molten salt to the ARC subsystem. A hot molten salt storage tank is configured to store the heated molten salt and pass the heated molten salt from the CR to the HRVG. The cold molten salt storage tank is configured to store the cold molten salt and pass the cold molten salt from the ADRC subsystem to the CR. The HTF pump is configured to pump the cold molten salt from the cold molten salt storage to the CR.

(8) In an aspect of the present disclosure, the ERC subsystem includes a first evaporator (E**1**), a steam turbine (T), a steam ejector, a first condenser (C**1**), a first throttling valve (TV**1**), and a first pump (P**1**). The first evaporator (E**1**) is configured to provide the first refrigerant effect with the first range of temperature by evaporating water flowing through the first evaporator (E**1**) into water vapor. The steam turbine (T) is configured to generate the electrical power based on the superheated steam from the HRVG of the SRC subsystem. The steam ejector is configured to generate a mixture by mixing an exhaust from the steam turbine (T) and the water vapor from the first evaporator (E**1**). The first condenser (C**1**) is configured to cool the mixture from the steam ejector into the water. TA first throttling valve (TV**1**) is configured to pass a part of the water from the first condenser (C**1**) to the first evaporator (E**1**) and reduce a pressure of the part of the water to a pressure of the first evaporator (E**1**). The first pump (P**1**) is configured to pump the other part of the water from the first condenser (C**1**) to the HRVG of the SRC subsystem.

(9) In an aspect of the present disclosure, the ARC subsystem includes a generator (G), a second condenser (C**2**), a second evaporator (E**2**), a second throttle valve (TV**2**), an absorber (A), a solution heat exchanger (SHX), a second pump (P**2**), a third throttling valve (TV**3**). The generator (G) is configured to pass the heated molten salt from the HRVG of the SRC subsystem to adsorption chambers of the ADRC subsystem; generate superheated water vapor refrigerant based on the heated molten salt flowing through the generator (G); and generate a strong solution based on a weak solution flowing into the generator (G). The second condenser (C**2**) is configured to generate saturated liquid refrigerant by cooling the superheated water vapor refrigerant from the generator (G). The second evaporator (E**2**) is configured to provide the second refrigeration effect with the second range of temperature by generating saturated water vapor based on the saturated liquid refrigerant from the second condenser (C**2**). The second throttle valve (TV**2**) is configured to pass the saturated liquid refrigerant from the second condenser (C**2**) to the second evaporator (E**2**) and reduce a pressure of the saturated liquid refrigerant to a pressure of the second evaporator (E**2**). The absorber (A) is configured to generate the weak solution by mixing the strong solution with the

saturated water vapor from the second evaporator (E**2**). The solution heat exchanger (SHX) is configured to pre-heat the weak solution from the absorber (A); pass the pre-heated weak solution to the generator (G); and pre-cool the strong solution from the generator (G). The third throttling valve (TV**3**) configured to pass the pre-cooled strong solution from the solution heat exchanger (SHX) to the absorber (A) and reduce a pressure of the pre-cooled strong solution to a pressure of the absorber (A).
(10) In an aspect of the present disclosure, the BRC subsystem includes a third evaporator (E**3**), a compressor (COM P), a third condenser (C**3**), a fourth throttling valve (TV**4**), and an ice storage system. The third evaporator (E**3**) is configured to provide the third refrigeration effect with the third range of temperature by evaporating liquid refrigerant flowing through the third evaporator (E**3**) into superheated refrigerant vapor. The compressor (COMP) is powered by the electrical power generated by the steam turbine of the ERC subsystem and is configured to generate high-pressure superheated refrigerant vapor by compressing the superheated refrigerant vapor from the third evaporator (E**3**). The third condenser (C**3**) is configured to cool the high-pressure superheated refrigerant vapor into the liquid refrigerant flowing to the third evaporator (E**3**). The fourth throttling valve (TV**4**) is configured to pass the liquid refrigerant from the third condenser (C**3**) to the third evaporator (E**3**) and reduce a pressure of the liquid refrigerant to a pressure of the third evaporator (E**3**). The ice storage system is configured to store the ice when there is no cooling demand and melt the ice when there is cooling demand.
(11) In an aspect of the present disclosure, the ADRC subsystem includes two adsorption chambers (AC-**1** and AC-**2**), a fourth evaporator (E**4**), and a fourth condenser (C**4**). The two adsorption chambers (AC-**1** and AC-**2**) are configured to perform as adsorber and desorber alternatively and respectively. The fourth evaporator (E**4**) is configured to provide the fourth refrigerant effect with the fourth range of temperature by evaporating water flowing through the four evaporator (E**4**) into water vapor flowing to a first adsorption chamber of the two adsorption chambers performing as the absorber. The fourth condenser (C**4**) is configured to cool the water vapor from a second adsorption chamber of the two adsorption chambers performing as the desorber. The heated molten salt is cooled through the two adsorption chambers and returns to the cold molten salt storage tank of the SRC subsystem.
(12) According to another aspect of the present disclosure, a system includes a solar-powered steam Rankine cycle (SRC) subsystem, an organic Rankine cycle (ORC) subsystem, an absorption refrigeration cycle (ARC) subsystem, a multi-effect desalination (MED) subsystem, a reverse osmosis (RO) desalination subsystem, and a hydrogen production subsystem. The SRC subsystem is configured to receive solar energy, convert the solar energy into thermal energy, and store the thermal energy. The ORC subsystem is configured to generate an electrical power based on the thermal energy. The ARC subsystem is configured to generate a refrigeration effect based on the thermal energy. The MED subsystem is configured to desalinate seawater to produce first desalinated water based on steam generated from the ORC subsystem. The RO desalination subsystem is configured to desalinate the seawater to produce second desalinated water based on the electrical power generated by the ORC subsystem. The hydrogen production subsystem is configured to produce hydrogen from the second desalinated water produced by the RO desalination subsystem.
(13) In an aspect of the present disclosure, the SRC subsystem includes a central receiver (CR), a heat recovery vapor generator (HRVG), a hot molten salt storage tank, a cold molten salt storage tank, and a heat transfer fluid (HTF) pump (P**1**). The central receiver (CR) is configured to receive the solar energy to heat cold molten salt flowing through the CR. The heat recovery vapor generator (HRVG) is configured to generate superheated steam for the ORC subsystem based on the heated molten salt and pass the heated molten salt to the ARC subsystem. The hot molten salt storage tank is configured to store the heated molten salt and pass the heated molten salt from the CR to the HRVG. The cold molten salt storage tank is configured to store the cold molten salt and

pass the cold molten salt from the ARC subsystem to the CR. The heat transfer fluid (HTF) pump (P**1**) is configured to pump the cold molten salt from the ARC subsystem to the cold molten salt storage tank.

(14) In an aspect of the present disclosure, the ORC subsystem includes a high-pressure turbine (HPT), a low-pressure turbine (LPT), a steam ejector (EJE), a first condenser (C**1**), a heat exchanger (HX), a third pump (P**3**), and a fourth pump (P**4**). The high-pressure turbine (HPT) is configured to generate a first part of the electrical power based on the superheated steam from the HRVG of the SRC subsystem. The low-pressure turbine (LPT) is configured to generate a second part of the electrical power based on the superheated steam from the HRVG of the SRC subsystem. The steam ejector (EJE) is configured to generate a mixture by mixing an exhaust from the high-pressure turbine (HPT) and vapor from the MED subsystem; and pass the mixture to the MED subsystem. The first condenser (C**1**) is configured to condense steam from a turbine (T) of the ORC subsystem into water. The heat exchanger (HX) is configured to generate condensate based on exhaust steam from the low-pressure turbine (LPT) and generate steam based on the water from the first condenser (C**1**). The third pump (P**3**) is configured to pump the condensate from the heat exchanger (HX) to the HRVG of the SRC subsystem. The fourth pump (P**4**) is configured to pump the cooled water from the first condenser (C**1**) to the heat exchanger (HX). The turbine (T) is configured to generate a third part of the electrical power based on the steam from the heat exchanger (HX).

(15) In an aspect of the present disclosure, the ARC subsystem includes a generator (G), a second condenser (C**2**), a first evaporator (E**1**), a first throttle valve (TV**1**), an absorber (A), a solution heat exchanger (SHX), a fifth pump (P**5**), and a second throttling valve (TV**2**). The generator (G) is configured to pass the heated molten salt from the HRVG to the cold molten salt storage tank of the SRC subsystem; generate superheated water vapor refrigerant based on the heated molten salt flowing through the generator (G); and generate a strong solution based on a weak solution flowing into the generator (G). The second condenser (C**2**) is configured to generate saturated liquid refrigerant by cooling the superheated water vapor refrigerant from the generator (G). The first evaporator (E**1**) is configured to provide the refrigeration effect by generating saturated water vapor based on the saturated liquid refrigerant from the second condenser (C**2**). The first throttle valve (TV**1**) is configured to pass the saturate liquid refrigerant from the second condenser (C**2**) to the first evaporator (E**1**) and reduce a pressure of the saturated liquid refrigerant to a pressure of the first evaporator (E**1**). The absorber (A) is configured to generate the weak solution by mixing the strong solution with the saturated water vapor from the first evaporator (E**1**). The solution heat exchanger (SHX) is configured to pre-heat the weak solution from the absorber (A); pass the pre-heated weak solution to the generator (G); and pre-cool the strong solution from the generator (G). The fifth pump (P**5**) is configured to pump the weak solution from the absorber (A) to the SHX. The second throttling valve (TV**2**) is configured to pass the pre-cooled strong solution from the SHX to the absorber (A) and reduce a pressure of the pre-cooled strong solution to a pressure of the absorber (A).

(16) In an aspect of the present disclosure, the MED subsystem includes a third condenser (C**3**) and a second pump (P**2**). The third condenser (C**3**) is configured to heat and forward the seawater to a first evaporation effect of a plurality of evaporation effects. In an aspect of the present disclosure, the plurality of evaporation effects is configured to produce the first desalinated water from the heated seawater, where the first evaporation effect is driven by the mixture from the steam ejector of the ORC subsystem. The second pump (P**2**) is configured to pump condensate from the first evaporation effect to the HRVG of the SRC subsystem.

(17) In an aspect of the present disclosure, the RO desalination subsystem includes a high-pressure pump (P**6**) configured to pump seawater to a semi-permeable RO membrane that demineralizes the seawater to produce the second desalinated water.

(18) In an aspect of the present disclosure, the hydrogen production subsystem includes a proton

exchange membrane (PEM) electrolyzer, a hydrogen compressor, a hydrogen storage tank, and an oxygen storage tank. The proton exchange membrane (PEM) electrolyzer is configured to produce hydrogen and oxygen through a water separation process of the second desalinated water from the RO membrane of the RO desalination subsystem. The hydrogen compressor is configured to compress and cool the hydrogen, the hydrogen storage tank is configured to store the compressed hydrogen, and the oxygen storage tank is configured to store the produced oxygen. The high-pressure pump (P**6**) is powered by the second part of the electrical power generated by the low-pressure turbine (LPT) of the ORC subsystem and both the PEM electrolyzer and the hydrogen compressor are powered by the third part of the electrical power generated by the turbine (T) of the ORC subsystem.

(19) According to yet another aspect of the present disclosure, a system includes a solar-powered steam Rankine cycle (SRC) subsystem, an absorption refrigeration cycle (ARC) subsystem, a multi-effect desalination (M ED) subsystem, a trans critical refrigeration cycle (TRC) subsystem, and a freezing desalination (FRD) subsystem. The SRC subsystem is configured to receive solar energy, convert the solar energy into thermal energy, and store the thermal energy. The ARC subsystem is configured to provide a first refrigeration effect with a first range of temperature based on the thermal energy. The MED subsystem is configured to desalinate seawater to produce first desalinated water based on the thermal energy and generate an electrical power based on the thermal energy. The TRC subsystem is configured to store ice when there is no cooling demand and provide a second refrigeration effect with a second range of temperature based on the electrical energy generated by the MED subsystem and the ice being melted. The FRD subsystem is configured to desalinate the seawater to produce second desalinated water based the ice in the TRC subsystem being melted.

(20) In an aspect of the present disclosure, the SRC subsystem includes a central receiver (CR), a heat recovery vapor generator (HRVG), a hot molten salt storage tank, a cold molten salt storage tank, and a heat transfer fluid (HTF) pump (P**1**). The CR is configured to receive the solar energy to heat cold molten salt flowing through the central receiver. The HRVG is configured to generate superheated steam for the MED subsystem based on the heated molten salt and pass the heated molten salt to the ARC subsystem. The hot molten salt storage tank is configured to store the heated molten salt and pass the heated molten salt from the CR to the HRVG. The cold molten salt storage tank is configured to store the cold molten salt and pass the cold molten salt from the ARC subsystem to the CR. The heat transfer fluid (HTF) pump (P**1**) is configured to pump the cold molten salt from the ARC subsystem to the cold molten salt storage tank.

(21) In an aspect of the present disclosure, the ARC subsystem includes a generator (G), a first condenser (C**1**), a first evaporator (E**1**), a first throttle valve (TV**1**), an absorber (A), a solution heat exchanger (SHX), a third pump (P**3**), and a third throttling valve (TV**3**). The generator (G) is configured to pass the heated molten salt from the HRVG to the cold molten salt storage tank of the SRC subsystem; generate superheated water vapor refrigerant based on the heated molten salt flowing through the generator (G); and generate a strong solution based on a weak solution flowing into the generator (G). The first condenser (C**1**) is configured to generate saturated liquid refrigerant by cooling the superheated water vapor refrigerant from the generator (G). The first evaporator (E**1**) is configured to provide the first refrigeration effect with the first range of temperature by generating saturated water vapor based on the saturated liquid refrigerant from the first condenser (C**1**). The first throttle valve (TV**1**) is configured to pass the saturate liquid refrigerant from the first condenser (C**1**) to the first evaporator (E**1**) and reduce a pressure of the saturated liquid refrigerant to a pressure of the first evaporator (E**1**). The absorber (A) is configured to generate the weak solution by mixing the strong solution with the saturated water vapor from the first evaporator (E**1**). The solution heat exchanger (SHX) is configured to pre-heat the weak solution from the absorber (A), pass the pre-heated weak solution to the generator (G), and pre-cool the strong solution from the generator (G). The third pump (P**3**) is configured to pump the weak

solution from the absorber (A) to the SHX. The third throttling valve (TV**3**) is configured to pass the pre-cooled strong solution from the SHX to the absorber (A) and reduce a pressure of the pre-cooled strong solution to a pressure of the absorber (A).
(22) In an aspect of the present disclosure, the MED subsystem includes a steam turbine (T), a steam ejector (EJE), a second condenser (C**2**), and a second pump (P**2**). The steam turbine (T) is configured to generate the electrical power based on the superheated steam from the HRVG of the SRC subsystem. The steam ejector (EJE) is configured to generate a mixture by mixing an exhaust from the steam turbine (T) and vapor from the MED subsystem and pass the mixture to the MED subsystem. The second condenser (C**2**) is configured to heat and forward seawater to a first evaporation effect of a plurality of evaporation effects. The plurality of evaporation effects is configured to produce fresh water from the heated seawater, where the first evaporation effect is driven by the mixture from the steam ejector (EJE). The second pump (P**2**) is configured to pump condensate from the first evaporation effect to the HRVG of the SRC subsystem.
(23) In an aspect of the present disclosure, the TRC subsystem includes a second evaporator (E**2**), an internal heat exchanger (IHX), a compressor (COM P), a third condenser (C**3**), a second throttling valve (TV**2**), and an ice storage system. The second evaporator (E**2**) is configured to provide the second refrigeration effect with the second range of temperature by generating refrigerant vapor. The internal heat exchanger (IHX) is configured to generate superheated refrigerant vapor by evaporating liquid droplets of the refrigerant vapor from the second evaporator (E**2**). The compressor (COM P) is powered by the electrical power generated from the steam turbine of the MED subsystem and is configured to generate high-pressure superheated refrigerant vapor by compressing the superheated refrigerant vapor from the internal heat exchanger (IHX). The third condenser (C**3**) is configured to cool the high-pressure superheated refrigerant vapor to saturated liquid refrigerant by melting ice in the FRD subsystem. The second throttling valve (TV**2**) is configured to pass the saturated liquid refrigerant to the second evaporator (E**2**) and reduce a pressure of the saturated liquid refrigerant to a pressure of the second evaporator (E**2**). The ice storage system is configured to build the ice when there is no cooling demand and melt the ice when there is cooling demand.
(24) In an aspect of the present disclosure, the FRD subsystem includes a washer configured to separate ice and brine that are transported from the ice storage tank of the TRC subsystem; and a fourth condenser (C**4**) configured to pre-cool seawater through a product fresh water steam from an ice melter and discharged brine steam from the washer.
(25) In an aspect of the present disclosure, the MED subsystem is configured to operate in one of parallel feed configuration, forward feed configuration, backward feed configuration, and parallel/cross flow feed configuration.
(26) These and other aspects of non-limiting embodiments of the present disclosure will become apparent to those skilled in the art upon review of the following description of specific non-limiting embodiments of the disclosure in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A better understanding of embodiments of the present disclosure (including alternatives and/or variations thereof) may be obtained with reference to the detailed description of the embodiments along with the following drawings, in which:
(2) FIG. **1**A-C illustrates an exemplary schematic diagram of a system, according to an aspect of the present disclosure;
(3) FIG. **2**A-C illustrates a forward feed configuration of the system, according to an aspect of the present disclosure;

(4) FIG. **3**A-C illustrates a parallel feed configuration of the system, according to an aspect of the present disclosure;
(5) FIG. **4**A-C illustrates a parallel/cross flow feed configuration of the system, according to an aspect of the present disclosure;
(6) FIG. **5**A-C illustrates an exemplary instrumentation diagram of the system, according to an aspect of the present disclosure;
(7) FIG. **6** illustrates an advanced control system for the system of FIG. **1**A-C through FIG. **5**A-C, according to an aspect of the present disclosure;
(8) FIG. **7**A-C illustrates a forward feed configuration of another system, according to an aspect of the present disclosure;
(9) FIG. **8**A-C illustrates a parallel feed configuration of the system of FIG. **7**A-C, according to an aspect of the present disclosure;
(10) FIG. **9**A-C illustrates a parallel/cross flow feed configuration of the system of FIG. **7**A-C, according to an aspect of the present disclosure;
(11) FIG. **10**A-C illustrates an exemplary instrumentation diagram of the system of FIG. **7**A-C, according to an aspect of the present disclosure; and
(12) FIG. **11** illustrates an advanced control system for the system of FIG. **7**A-C through FIG. **10**A-C, according to an aspect of the present disclosure.

DETAILED DESCRIPTION

(13) In the following description, it is understood that other embodiments may be utilized, and structural and operational changes may be made without departure from the scope of the present embodiments disclosed herein.

(14) Reference will now be made in detail to specific embodiments or features, examples of which are illustrated in the accompanying drawings. Wherever possible, corresponding, or similar reference numbers will be used throughout the drawings to refer to the same or corresponding parts. Moreover, references to various elements described herein, are made collectively or individually when there may be more than one element of the same type. However, such references are merely exemplary in nature. It may be noted that any reference to elements in the singular may also be construed to relate to the plural and vice-versa without limiting the scope of the disclosure to the exact number or type of such elements unless set forth explicitly in the appended claims.

(15) In the drawings, like reference numerals designate identical or corresponding parts throughout the several views. Further, as used herein, the words “a,” “an” and the like generally carry a meaning of “one or more,” unless stated otherwise.

(16) Furthermore, the terms “approximately,” “approximate,” “about,” and similar terms generally refer to ranges that include the identified value within a margin of 20%, 10%, or preferably 5%, and any values therebetween.

(17) Aspects of the present disclosure are directed to a multi-effect thermodynamic cycle driven by solar power, which may produce a refrigeration output at different levels of temperature. The multi-effect thermodynamic cycle may provide cooling for air conditioning (10° C. to 20° C.), refrigeration for food preservation (2° C. to 9° C.), for vegetable and meat storage (1° C. to 5° C.), freezing for pharmaceutical plants and vaccine (−60° C. to −80° C.).

(18) FIG. **1**A-C illustrates an exemplary schematic diagram of a system **100**, according to an embodiment of the present disclosure. The system **100** includes a solar-powered steam Rankine cycle (SRC) subsystem **102** (hereinafter referred to as “the SRC subsystem **102**”) configured to receive solar energy, convert the solar energy into thermal energy, and store the thermal energy. The SRC **102** includes a central receiver (CR) **104** fixed on a solar tower **105**, a heat recovery vapor generator (HRVG) **106**, a hot molten salt storage tank **108**, a cold molten salt storage tank **110**, and a heat transfer fluid (HTF) pump **112**. Solar radiation incident on a heliostat field **114** is reflected to the CR **104**. In an aspect of the present disclosure, the CR **104** is configured to: (a) allow flow of cold molten salt from the cold molten storage tank **110** therethrough and (b) receive the reflected

solar radiation to heat the cold molten salt flowing through the CR **104**.
(19) The hot molten salt storage tank **108** is connected between the CR **104** and the HRVG **106**. As shown in FIG. **1**A-C, the hot molten salt storage tank **108** is connected to the CR **104** via a flow path **116** and to the HRVG **106** via a flow path **118**. The heated molten salt is pumped from the CR **104** to the hot molten salt storage tank **108** via the flow path **116**. The hot molten salt storage tank **108** is configured to store the heated molten salt and pass the heated molten salt to the HRVG **106** via the flow path **118**. At the HRVG **106**, configuration of the HRVG **106** allows transfer of heat between the heated molten salt and water flowing therethrough. In an aspect of the present disclosure, the HRVG **106** is configured to generate superheated steam based on the heated molten salt. The HRVG **106** is connected to an ejector refrigeration cycle (ERC) subsystem **120** (hereinafter referred to as “the ERC subsystem **120**”) via a flow path **122** and to an absorption refrigeration cycle (ARC) subsystem **124** (hereinafter referred to as “the ARC subsystem **124**”) via a flow path **126**. The ERC subsystem **120** receives the generated superheated steam from the HRVG **106** via the flow path **122** and the heated molten salt is passed by the HRVG **104** to the ARC subsystem **124** via the flow path **126**.
(20) In an aspect of the present disclosure, The ERC subsystem **120** includes a first evaporator (E**1**) **128**, a steam turbine (T) **130**, a steam ejector **132**, a first condenser (C**1**) **134**, a first throttling valve (TV**1**) **136**, and a first pump (P**1**) **138**. The ERC subsystem **120** is configured to generate an electrical power based on the thermal energy received from the
(21) SRC subsystem **102** and provide a first refrigeration effect with a first range of temperature based on the thermal energy. The superheated steam generated by the HRVG **106** is supplied to the steam turbine (T) **130** via a flow path **140**. The steam turbine (T) **130** is configured to generate the electrical power based on an expansion of the superheated steam received from the HRVG **106**. The first evaporator (E**1**) **128** is configured to provide the first refrigerant effect with the first range of temperature by evaporating water flowing through the first evaporator (E**1**) **128** into water vapor.
(22) An exit path **142** of the steam turbine (T) **130** is connected to the steam ejector **132**. An exhaust from the steam turbine (T) **130** passes through the steam ejector **132** via the exit path **142** and creates a high vacuum pressure that draws the water vapor generated by the first evaporator (E**1**) **128**, thereby aiding the first refrigerant effect of the ERC subsystem **120**. The steam ejector **132** is configured to generate a mixture by mixing the exhaust from the steam turbine (T) **130** and the water vapor from the first evaporator (E**1**) **128**. The mixture flows to the first condenser (C**1**) **134** via a flow path **144**. The first condenser (C**1**) **134** is configured to cool the mixture into the water. In an aspect of the present disclosure, a condenser fluid may be cooled by cold water supplied from a cooling tower **145**. An outlet of the first condenser (C**1**) **134** branches to join the flow path **122** on one side and the first throttling valve (TV**1**) **136** on the other side as shown in FIG. **1**A-C.
(23) The first throttling valve (TV**1**) **136** is configured to pass a first volume of the water from the first condenser (C**1**) **134** to the first evaporator (E**1**) **128** and reduce a pressure of the part of the water to a pressure of the first evaporator (E**1**) **128**. The first pump (P**1**) **138** is configured to pump a second volume of the water from the first condenser (C**1**) **134** to the HRVG **106** of the SRC subsystem **102**. The water entering into the HRVG **106** is heated by the molten salt flowing through the HRVG **106** to convert into the superheated steam which further flows into the steam turbine (T).
(24) In an aspect of the present disclosure, The ARC subsystem **124** is configured to provide a second refrigeration effect with a second range of temperature based on the thermal energy. The ARC subsystem **124** includes a generator (G) **146**, a second condenser (C**2**) **148**, a second throttle valve (TV**2**) **150**, a second evaporator (E**2**) **152**, an absorber (A) **154**, a third throttling valve (TV**3**) **156**, a solution heat exchanger (SHX) **158**, and a second pump (P**2**) **160**. The generator (G) **146** receives the heated molten salt from the HRVG **106** via the flow path **126**. The generator (G) **146** is configured to pass the heated molten salt from the HRVG **106** to adsorption chambers of an

adsorption refrigeration cycle (ADRC) subsystem **162** (hereinafter referred to as the "ADRC **162** subsystem"). The generator (G) **146** is further configured to generate superheated water vapor refrigerant based on the heated molten salt flowing therethrough and generate a strong solution based on a weak solution flowing therein. The second condenser (C**2**) **148** is connected to the generator (G) **146** via a flow path **164** and is configured to generate saturated liquid refrigerant by cooling the superheated water vapor refrigerant received from the generator (G) **146**.

(25) Heat rejected by the superheated water vapor refrigerant in the second condenser (C**2**) **148** is cooled by water supplied from the cooling tower **145**. The second evaporator (E**2**) **152** is connected to the second condenser (C**2**) **148** via a flow path **168**, where the second throttle valve (TV**2**) **150** is connected on the flow path **168**. The second throttle valve (TV**2**) **150** is configured to pass the saturated liquid refrigerant from the second condenser (C**2**) **148** to the second evaporator (E**2**) **152** and reduce a pressure of the saturated liquid refrigerant to a pressure of the second evaporator (E**2**) **152**. The second evaporator (E**2**) **152** is configured to provide the second refrigeration effect with the second range of temperature by generating saturated water vapor based on the saturated liquid refrigerant from the second condenser (C**2**) **148**.

(26) The saturated water vapor absorbs heat from the second evaporator (E**2**) and flows to the absorber (A) **154** that is connected to the second evaporator (E**2**) **152** via a flow path **170**. The absorber (A) **154** is configured to generate the weak solution by mixing the strong solution with the saturated water vapor. The weak solution is pumped from the absorber (A) **154** to the solution heat exchanger (SHX) **158** by the second pump (P**2**) **160** via a flow path **172**. The solution heat exchanger (SHX) **158** is configured to pre-heat the weak solution from the absorber (A) **154**, pass the pre-heated weak solution to the generator (G) **146** via a flow path **174**, and pre-cool the strong solution from the generator (G) **146**. In one aspect, the strong solution received by the solution heat exchanger (SHX) **158** from the generator (G) **146** may be LiBr—H.sub.2O solution. The third throttling valve (TV**3**) **156** is configured to reduce a pressure of the pre-cooled strong solution from the solution heat exchanger (SHX) **158** to a pressure of the absorber (A) **154** and pass the pre-cooled strong solution to the absorber (A) **154**. In an aspect of the present disclosure, temperature of the absorber (A) **154** may be maintained at 35° C. and at a pressure equal to a pressure of the second evaporator (E**2**) **152**.

(27) The system **100** also includes a brine refrigeration cycle (BRC) subsystem **176** (hereinafter referred to as "the BRC subsystem **176**"). The BRC subsystem **176** is configured to build and store ice when there is no cooling demand and provide a third refrigeration effect with a third range of temperature based on the electrical power generated by the ERC subsystem **120** and the ice being melted. In an aspect of the present disclosure, the BRC subsystem **176** includes a third evaporator (E**3**) **177**, a compressor (COMP) **178**, a third condenser (C**3**) **179**, a fourth throttling valve (TV**4**) **180**, and an ice storage system **181**. The third evaporator (E**3**) **177** is configured to provide the third refrigeration effect with the third range of temperature by evaporating liquid refrigerant flowing through the third evaporator (E**3**) **177** into superheated refrigerant vapor. In an aspect of the present disclosure, N.sub.2O may be used in the BRC subsystem **176** to provide a very low-temperature range and the third refrigeration effect that can be used for deep freezing applications in a range of −50° C. to −80° C. The compressor (COMP) **178** is powered by the electrical power generated from the steam turbine (T) **130** of the ERC subsystem **120** and is connected to the third evaporator (E**3**) **177** via a flow path **182**. The compressor (COMP) **178** is configured to generate high-pressure superheated refrigerant vapor by compressing the superheated refrigerant vapor from the third evaporator (E**3**) **177**.

(28) The third condenser (C**3**) **179** is connected to the compressor (COMP) **178** via a flow path **183**. The third condenser (C**3**) **179** is configured to receive the high-pressure superheated refrigerant vapor (such as the superheated N **20** refrigerant vapor) and cool the high-pressure superheated refrigerant vapor into the liquid refrigerant flowing to the third evaporator (E**3**) **177**. In an aspect of the present disclosure, the third condenser (C**3**) **179** may be cooled by the water

supplied from the cooling tower **145**. The water for cooling may be circulated between the cooling tower **145** and the third condenser (C**3**) **179** via a condenser pump (CP) **184**. The fourth throttling valve (TV**4**) **180** is configured to reduce a pressure of the liquid refrigerant to a pressure of the third evaporator (E**3**) **177** and pass the liquid refrigerant from the third condenser (C**3**) **179** to the third evaporator (E**3**) **177**. The third refrigeration effect in the third evaporator (E**3**) **177** is achieved by evaporating N.sub.2O refrigerant.
(29) The ice storage system **181** is configured to build the ice when there is no cooling demand and melt the ice when there is cooling demand. In an aspect of the present disclosure, the stored thermal energy of the ice storage system **181** may be used to meet changing cooling load demands from a state "S**1**" to a state "S**2**" (see FIG. **1**A-C). During peak-time and in case of grid outages, the stored thermal energy in the ice storage system **181** from a state "S**3**" to a state "S**4**" will continue to cool a building adding an increased level of redundancy. The ice will be melted when there is cooling demand and a secondary glycol pump (SGP) is started. A melt-out cycle may begin as the hot water comes from the building (corresponding to the state "S**1**") and passes through a heat exchanger (HX) to get converted to cold chilled water (corresponding to the state "S**2**") that circulates through a tertiary glycol pump (TGP) to meet a required cooling load. It may be observed that there is no direct contact between the melting ice and the hot return water which is cooled indirectly in the heat exchanger (HX) by melting ice.
(30) Table 1 below illustrates operation modes of the BRC subsystem **176** with the ice storage system **181**. The modes are Cooling-Ice with BRC, Cooling-BRC only, Cooling-Ice Only, Ice-Build, and Ice-Build with Cooling.
(31) TABLE-US-00001 TABLE 1 Sequence of Operation of Brine Refrigeration and Ice Storage System Operation Mode BRC PPG SGP MV-1 MV-2 Cooling-Ice with BRC On On On Variable Y1-Y2 Cooling-BRC Only On On On X1-X3 Y1-Y2 Cooling-Ice Only Off On On Variable Y1-Y2 Ice Build On On Off X1-X2 Y1-Y3 Ice Build with Cooling On On On X1-X2 Variable
(32) In an aspect, a sequence of operation and modulation of motorized valves MV-**1** and MV-**2** (see FIG. **1**A-C) depends on a temperature at the state "S**1**". The BRC subsystem **176** with ice storage may operate based on the modes provided in the Table-1. A combined system (that is, the BRC subsystem **176** with ice storage) may provide cooling through BRC subsystem **176** only. Therefore, in this mode, the BRC, PGP, and SGP shall be ON, and a circulation fluid is passed through three ports in MV-**1** as the chilled water will bypass through the state "S**4**" and two ports port in MV-**2** where the chilled water will pass to the SGP and then to the HX.
(33) In an aspect of the present disclosure, the system **100** may provide cooling through the BRC subsystem **176** and the stored ice. In this mode, the BRC, PGP, and SGP shall be ON, and the circulation of the fluid is modulated in MV-**1** to balance the load from the BRC subsystem **176** and Ice storage and allow the fluid to pass through two ports in MV-**2**.
(34) In another aspect of the present disclosure, cooling may be provided through the ice storage system **181**. In this mode, the BRC subsystem **176** is switched OFF, and the PGP and the SGP shall be ON while the circulation of the fluid is allowed through the three ports in MV-**1** and the two ports in MV-**2**.
(35) In yet another aspect of the present disclosure, the combined BRC subsystem **176** with ice storage may generate and store only during the night and part-load operation. In this mode, the BRC subsystem **176** and the PGP shall be running and the pump SGP is switched OFF. The circulation of the fluid is allowed through the two ports in MV-**1** and three ports in MV-**2** as the chilled water is passed to PGP and then returned to third evaporator (E**3**) **177**. In yet another embodiment, the combined BRC subsystem **176** with ice storage may generate and store and provide cooling at the same time. In this mode, the BRC, the PGP, and the SGP shall be running, and the circulation of the fluid is allowed through the two ports in MV-**1** and modulated in MV-**2** according to required cooling demand.
(36) According to an aspect of the present disclosure, working fluids selected for the system **100**

are environmentally safe and show excellent thermodynamic properties. The working fluids are chemically stable, and do not show a toxic, explosive, or corrosive effect. Additionally, the working fluids exhibit zero ozone depletion potential (ODP), very low global warming potential (GWP), and are available at a low initial and operational cost.
(37) TABLE-US-00002 TABLE 2 Multi-Effect Refrigeration Cycle Working Fluids. Temper- ature Cycle Working Fluid Range (° C.) Power Cycle 1. Steam Rankine cycle Molten Salt (a mixture of 290~565 With Solar tower sodium nitrate & potassium (SRC). nitrate). 2. Steam Rankine cycle Water (H.sub.2O). with HRVG. Multi-Effect Refrigeration Cycle 1. Ejector Refrigeration Water (H.sub.2O). 1-5 Cycle (ERC) 2. Brine Refrigeration Nitrous Oxide (N.sub.2O). −50 to −80 cycle (BRC) 3. Absorption Lithium bromide is the 10 to 20 Refrigeration Cycle absorbent, and water is the (ARC) refrigerant (LiBr—H.sub.2O). 4. Adsorption Zeolite 13X/CaCl.sub.2 2 to 9 Refrigeration Cycle (ADRC)
(38) In an aspect of the present disclosure, In an aspect of the present disclosure, the ADRC subsystem **162** includes two adsorption chambers AC-**1** and AC-**2**, a fourth evaporator (E**4**) **185**, and a fourth condenser (C**4**) **186**. The ADRC subsystem **162** is configured to provide a fourth refrigeration effect with a fourth range of temperature based on the thermal energy. As described earlier, the generator (G) **146** is configured to pass the heated molten salt from the HRVG **106** of the SRC subsystem **102** to the adsorption chambers of the ADRC subsystem **162**. Specifically, the heated molten salt flows from the generator (G) **146** to the adsorption chamber AC-**1** via flow path **187** to regenerate sorbent. Subsequently, the molten salt flows to the cold molten storage tank **110** via a flow path **188**. The cold molten storage tank **110** is configured to store the cold molten salt and pass the cold molten salt to the CR **104**. The HTF pump **112** is configured to pump the cold molten salt from the cold molten salt storage tank **110** to the CR **104**. As described above, the cold molten salt is heated by the solar radiation incident on the CR **104**. Solar thermal energy is stored in the two tanks to maintain continuous operation when solar energy is not available.
(39) In an aspect, two adsorption chambers AC-**1** and AC-**2** are configured to perform as adsorber and desorber alternatively. For example, sorption elements in adsorption chambers AC-**1** and AC-**2** may function in a first stage as adsorber and desorber and will alternate cyclically in a second stage of operation. The fourth evaporator (E**4**) **185** is configured to provide the fourth refrigerant effect with the fourth range of temperature by evaporating water flowing therethrough into water vapor flowing to a first adsorption chamber AC-**1** of the two adsorption chambers performing as the absorber. The fourth condenser (C**4**) **186** is configured to cool the water vapor from a second adsorption chamber AC-**2** of the two adsorption chambers performing as the desorber. In an aspect, the heated molten salt is cooled through the two adsorption chambers and returns to the cold molten salt storage tank **110** of the SRC subsystem **102**.
(40) In an aspect of the present disclosure, the first adsorption chamber AC-**1** may be cooled by water from the cooling tower **145**. In an example, Zeolite may be used as the sorbent to absorb the water vapor generated by the fourth evaporator (E**4**) **185**. The second adsorption chamber AC-**2** may be configured to heat the hot water flowing from the generator (G) **146** and resulting water vapor may flow to the fourth condenser (C**4**) **186**. Condensed water associated with high pressure flows to the fourth evaporator (E**4**) **185** after attaining a pressure equal to that of a pressure of the fourth evaporator (E**4**) **185**. In an aspect of the present disclosure, a system of nozzles may spray the water onto tubes of chilled water at the fourth evaporator (E**4**) **185**.
(41) Simultaneously, water vapor flowing from the fourth evaporator (E**4**) **185** to the first adsorption chamber AC-**1** passes through an open valve GV**1**. In order to ensure the flow of water vapor towards the first adsorption chamber AC-**1**, the pressure inside the first adsorption chamber AC-**1** needs to be lower than that of the fourth evaporator (E**4**) **185**. Therefore, in some embodiments, the first adsorption chamber AC-**1** may be pre-cooled by cold water supplied from the cooling water **145**. Such cooling may be required to remove the heat added to the first adsorption chamber AC-**1** by the adsorption process. When the adsorbent in the first adsorption

chamber AC-**1** is saturated with the water vapor, function of the first adsorption chamber AC-**1** is switched.

(42) In a switched process, the first adsorption chamber AC-**1** and the second adsorption chamber AC-**2** function as desorber and adsorber, respectively. In the switched process, valves GV**1** and GV**4** are closed, and valves GV**2** and GV**3** are open. However, pressure in each of the first adsorption chamber AC-**1** and the second adsorption chamber AC-**2** may need to be modified prior to the switch over. For the purpose, the second adsorption chamber AC-**2** is cooled by a coolant and the first adsorption chamber AC-**1** is heated through the hot molten salt from the SRC subsystem **102**, and all valves are actuated to closed positions. The heated molten salt is cooled through the two adsorption chambers AC-**1** and AC-**2** and returns to the cold molten salt storage tank **110** of the SRC subsystem **102**.

(43) To this end, the system **100** represents an integration of multi-refrigeration cycles driven by one source of energy (such as the solar energy) to exploit solar energy effectively and to produce a cooling effect in different magnitude and temperature ranges for air conditioning application, refrigeration, and deep freezing, in an efficient and environment-friendly manner. It is apparent from the above description that a high-temperature solar energy source like a solar tower is utilized to produce power using high-temperature availability while supplying the required thermal energy to applications that require low or medium temperature sources, like absorption and adsorption cycle. Owing to the storage of the thermal energy, requirement of any auxiliary energy source is eliminated. As such, the cooling effect may be achieved during absence of the solar energy. The system **100** may be capable of meeting cooling demand for following application and temperature ranges: (a) food preservation (2-9° C.) by the ADRC subsystem **162**; (b) air conditioning application (10-20° C.) by the ARC subsystem **124**; (c) deep freezing for vaccine preservation (−50° C. to −80° C.) by the BRC subsystem **176**; and (d) vegetable and meat storage (1-5° C.) by the ERC subsystem **120**.

(44) The system **100** may be implemented at any existing facility where a maximum cooling load is much greater than an average load with limited electric power availability. With the presence of the ice storage system **181**, the system **100** may be effective for process loads that are not well-matched with an available energy source. As such, extra cooling capacity may be addressed by the ice storage system **181** based on a cooling demand. Additionally, the ice storage system **181** is effective for systems with time-dependent energy costs. The ice will be produced during low electricity tariff period (that is during night-time) and will be stored for usage throughout a day. The system **100** eliminates use of a generator required for backup energy as the ice storage system **181** is capable of storing the required thermal energy instead of using an electrical power generator.

(45) A combination of the ice storage system **181** and the use of solar energy for air cooling purposes renders the system **100** unique in the field of air-conditioning and refrigeration. The ice storage system **181** may aid in reducing energy consumption by the system **100** by taking advantage of lower energy prices and low ambient conditions to produce refrigeration effect with a high-energy performance by using electricity at night instead of producing cooling at peak daytime hours. A cooling production at night leads to large energy savings and may lower peak electrical demand by 50%. Additionally, the ice storage system **181** leads to low temperature of chilled water which may need smaller pumps, thereby further reducing the power consumption by the system **100**.

(46) The ice storage system **181** also needs a lower flow rate due to a large temperature gradient, such as a range of 10° C. instead of traditional 5.5° C. Such reduction in water flow rate leads to reduce component size of the system **100**, pumping power, and initial cost significantly.

(47) In an aspect of the present disclosure, an option of integrating waste energy heat source may be provided albeit with minor modifications to the system **100** to achieve continuous operation in case of long periods of low solar radiation, such as during cloudy winter days.

(48) In an aspect of the present disclosure, the SRC subsystem **102** may be integrated with an

electric power generator (see FIG. **2**A-C) that is connected to a utility grid (see FIG. **2**A-C). In cases where the electrical energy generated by the steam turbine (T) **130** is more than a load required by the BRC subsystem **176**, the extra electrical energy may be exported to the utility grid. In an aspect of the present disclosure, a control panel (not shown) may be implemented in the system **100** to determine availability of the excess electrical energy which may be routed to the utility grid.
(49) In an aspect of the present disclosure, a control valve may be provided at the HRVG **106** with a bypass line connected to the generator (G) **146**. Such arrangement may allow modulation of the flow of molten salt depending on the cooling demand at the ARC subsystem **124** and the A DRC subsystem **162** compared to the cooling demand at the ERC subsystem **120** and the BRC subsystem **176**. Thus, when the cooling demand is low at the ERC subsystem **120** and the BRC subsystem **176**, a part of the molten salt may be bypassed directly to the ARC subsystem **124** and the ADRC subsystem **162** to increase an output cooling effect.
(50) The system **100** may reduce electrical distribution therein, such as transformers, motor control center, power panel, switchgear, switchboard, power cables, wire sizes, and soft starter panels.
(51) FIG. **2**A-C illustrates an exemplary schematic diagram of a system **200**, according to an embodiment of the present disclosure. The system **200** includes a solar-powered steam Rankine cycle (SRC) subsystem **201** (hereinafter referred to as “the SRC subsystem **201**”) configured to receive solar energy, convert the solar energy into thermal energy, and store the thermal energy. The system **200** also includes an organic Rankine cycle (ORC) subsystem **202** (hereinafter referred to as “the ORC subsystem **202**”) configured to generate an electrical power based on the thermal energy, an absorption refrigeration cycle (ARC) subsystem **203** (hereinafter referred to as “the ARC subsystem **203**”) configured to generate a refrigeration effect based on the thermal energy, a multi-effect desalination (MED) subsystem **204** (hereinafter referred to as “the MED subsystem **204**”) configured to desalinate seawater to produce first desalinated water based on steam generated from the ORC subsystem **202**, a reverse osmosis (RO) desalination subsystem **205** (hereinafter referred to as “the RO subsystem **205**”) configured to desalinate the seawater to produce second desalinated water based on the electrical power generated by the ORC subsystem **202**, and a hydrogen production subsystem **206** configured to produce hydrogen from the second desalinated water produced by the RO desalination subsystem **205**.
(52) In an aspect, the SRC subsystem **201** includes a central receiver (CR) **207**, a heat recovery vapor generator (HRVG) **208**, a hot molten salt storage tank **209**, a cold molten salt storage tank **210**, and a heat transfer fluid (HTF) pump (P**1**) **211**. The CR **207** is configured to receive the solar energy to heat cold molten salt flowing therethrough. The heat recovery vapor generator (HRVG) **208** is configured to generate superheated steam for the ORC subsystem **202** based on the heated molten salt and pass the heated molten salt to the ARC subsystem **203**. The hot molten salt storage tank **209** is configured to store the heated molten salt and pass the heated molten salt from the CR **207** to the HRVG **208**. The cold molten salt storage tank **210** is configured to store the cold molten salt and pass the cold molten salt from the ARC subsystem **203** to the CR **207**. The heat transfer fluid (HTF) pump (P**1**) **211** is configured to pump the cold molten salt from the ARC subsystem **203** to the cold molten salt storage tank **210**.
(53) In an aspect of the present disclosure, the ORC subsystem **202** includes a high-pressure turbine (HPT) **212**, a low-pressure turbine (LPT) **213**, a steam ejector (EJE) **214**, a first condenser (C**1**) **215**, a heat exchanger (HX) **216**, a third pump (P**3**) **217**, and a fourth pump (P**4**) **218**. The high-pressure turbine (HPT) **212** is configured to generate a first part of the electrical power based on the superheated steam from the HRVG **208** of the SRC subsystem **201**. The low-pressure turbine (LPT) **213** is configured to generate a second part of the electrical power based on the superheated steam from the HRVG **208**. The steam ejector (EJE) **214** is configured to generate a mixture by mixing an exhaust from the high-pressure turbine (HPT) **212** and vapor from the MED subsystem **204**; and pass the mixture to the MED subsystem **204**. The first condenser (C**1**) **215** is configured

to condense steam from a turbine (T) **219** of the ORC subsystem **202** into water. The heat exchanger (HX) **216** is configured to generate condensate based on exhaust steam from the low-pressure turbine (LPT) **213** and generate steam based on the water from the first condenser (C**1**) **215**. The third pump (P**3**) **217** is configured to pump the condensate from the heat exchanger (HX) **216** to the HRVG **208** of the SRC subsystem **201**. The fourth pump (P**4**) **218** is configured to pump the cooled water from the first condenser (C**1**) **215** to the heat exchanger (HX) **216**. The turbine (T) **219** is configured to generate a third part of the electrical power based on the steam from the heat exchanger (HX) **216**.

(54) In an aspect of the present disclosure, the ARC subsystem **203** includes a generator (G) **220**, a second condenser (C**2**) **221**, a first evaporator (E**1**) **222**, a first throttle valve (TV**1**) **223**, an absorber (A) **224**, a solution heat exchanger (SHX) **225**, a fifth pump (P**5**) **226**, and a second throttling valve (TV**2**) **227**. The generator (G) **220** is configured to pass the heated molten salt from the HRVG **208** to the cold molten salt storage tank **210** of the SRC subsystem **201**; generate superheated water vapor refrigerant based on the heated molten salt flowing through the generator (G) **220**; and generate a strong solution based on a weak solution flowing into the generator (G) **220**. The second condenser (C**2**) **221** is configured to generate saturated liquid refrigerant by cooling the superheated water vapor refrigerant from the generator (G) **220**. The first evaporator (E**1**) **222** is configured to provide the refrigeration effect by generating saturated water vapor based on the saturated liquid refrigerant from the second condenser (C**2**) **221**. The first throttle valve (TV**1**) **223** is configured to pass the saturate liquid refrigerant from the second condenser (C**2**) **221** to the first evaporator (E**1**) **222** and reduce a pressure of the saturated liquid refrigerant to a pressure of the first evaporator (E**1**) **222**. The absorber (A) **224** is configured to generate the weak solution by mixing the strong solution with the saturated water vapor from the first evaporator (E**1**) **222**. The solution heat exchanger (SHX) **225** is configured to pre-heat the weak solution from the absorber (A) **224**; pass the pre-heated weak solution to the generator (G) **220**; and pre-cool the strong solution from the generator (G) **220**. The fifth pump (P**5**) **226** is configured to pump the weak solution from the absorber (A) **224** to the solution heat exchanger (SHX) **225**. The second throttling valve (TV**2**) **227** is configured to pass the pre-cooled strong solution from the solution heat exchanger (SHX) **225** to the absorber (A) **224** and reduce a pressure of the pre-cooled strong solution to a pressure of the absorber (A) **224**.

(55) In an aspect of the present disclosure, the MED subsystem **204** includes a third condenser (C**3**) **228** and a second pump (P**2**) **229**. The third condenser (C**3**) **228** is configured to heat and forward the seawater to a first evaporation effect of a plurality of evaporation effects. In an aspect of the present disclosure, the plurality of evaporation effects is configured to produce the first desalinated water from the heated seawater, where the first evaporation effect is driven by the mixture from the steam ejector of the ORC subsystem **202**. The second pump (P**2**) **229** is configured to pump condensate from the first evaporation effect to the HRVG **208**.

(56) In an aspect of the present disclosure, the RO subsystem **205** includes a high-pressure pump (P**6**) **230** configured to pump seawater to a semi-permeable RO membrane **231** that demineralizes the seawater to produce the second desalinated water.

(57) In an aspect of the present disclosure, the hydrogen production subsystem **206** includes a proton exchange membrane (PEM) electrolyzer **232**, a hydrogen compressor **233**, a hydrogen storage tank **234**, and an oxygen storage tank **235**. The proton exchange membrane (PEM) electrolyzer **232** is configured to produce hydrogen and oxygen through a water separation process of the second desalinated water from the semi-permeable RO membrane **231** of the RO subsystem **205**. The hydrogen compressor **233** is configured to compress and cool the hydrogen, the hydrogen storage tank **234** is configured to store the compressed hydrogen, and the oxygen storage tank **235** is configured to store the produced oxygen. The high-pressure pump (P**6**) **230** is powered by the second part of the electrical power generated by the low-pressure turbine (LPT) **213** of the ORC subsystem **202** and both the PEM electrolyzer **232** and the hydrogen compressor **233** are powered

by the third part of the electrical power generated by the turbine (T) **219** of the ORC subsystem **202**.

(58) In operation of the system **200**, solar radiation incident on a heliostat field **236** is reflected to the CR **207** that is fixed on a solar tower **237**. Thermal energy from the reflected solar radiation is used to heat the cold molten salt contained in the CR **207**. In an aspect of the present disclosure, the cold molten salt may be heated to a temperature of about 565° C. In an aspect of the present disclosure, the CR **207** may be coated with a selective coating to minimize reflectivity, thereby improving absorption performance. The heated molten salt flows from the CR **207** to the hot molten salt storage tank **209** via a flow path **238**. The heated molten salt may be stored by the hot molten salt storage tank **209** and may be subsequently passed to the HRVG **208** via a flow path **239**. Heat transfer between the heated molten salt and the water occurs at the HRVG **208**, where superheated steam is generated for the ORC subsystem **202**. The generated superheated steam is used to drive the high-pressure turbine (HPT) **212**. The molten salt, after the heat transfer at the HRVG **208**, flows to the generator (G) **220** via a flow path **240**. The molten salt is then pumped to the cold molten salt storage tank **210** by the HTF pump (P**1**) **211**. Further, the molten salt at a temperature of about 350° C. is used for process heating **241**, where the temperature of the molten salt is reduced is about 290° C. before the molten salt is allowed to flow to the CR **207** via a flow path **242**. Solar energy is stored in two tanks to aid continuous operation of the system **200** during unavailability of the solar energy.

(59) The high-pressure turbine (HPT) **212** and the low-pressure turbine (LPT) **213** are configured to convert the available thermal energy to the first part of the electrical power and the second part of the electrical power, respectively. The superheated steam generated at the HRVG **208** is expanded in the high-pressure turbine (HPT) **212**. An exhaust from the high-pressure turbine (HPT) **212** passes to the steam ejector (EJE) **214** and creates a high vacuum pressure that causes drawing of vapor from the MED subsystem **204**. The steam ejector (EJE) **214** generates the mixture by mixing the vapor from the MED subsystem **204** and the exhaust from the high-pressure turbine (HPT) **212** and passes the mixture to the first evaporation effect (Effect-**1**) **244** of the MED subsystem **204**. FIG. **2**A-C also illustrates a forward feed configuration of the MED subsystem **204**, among other configurations.

(60) At the MED subsystem **204**, feed seawater **243** enters the third condenser (C**3**) **228** to be pre-heated. The feed seawater **243** then flows to the first evaporation effect (Effect-**1**) **244** via a flow path **245** and is sprayed in the first evaporation effect (Effect-**1**) **244** as shown in FIG. **2**A-C. The mixture from the steam ejector (EJE) **214** drives the evaporation of the feed seawater **243** in the first evaporation effect (Effect-**1**) **244**. Upon providing required energy for the evaporation, the mixture is condensed, and the condensate is pumped to the HRVG **208** by the second pump (P**2**) **229**. The steam ejector (EJE) **214** is actuated by the exhaust from the high-pressure turbine (HPT) **212** and entrains a portion of formed vapor from the second evaporation effect (Effect-**2**) **246**. The steam ejector (EJE) **214** is configured to compress the entrained vapor by the exhaust from the high-pressure turbine (HPT) **212** to reach a desired temperature and pressure along a flow path **247**.

(61) Further, vapor formed in the first evaporation effect (Effect-**1**) **244** from the evaporation of the feed seawater **243** is supplied to the second evaporation effect (Effect-**2**) **246** to drive the evaporation therein. Remainder of the feed seawater **243** which is not evaporated collects a bottom of the first evaporation effect (Effect-**1**) **244** and is pumped to the second evaporation effect (Effect-**2**) **246** as shown in FIG. **2**A-C. Similarly, vapor formed in the second evaporation effect (Effect-**2**) **246** is used to drive the evaporation in the third evaporation effect (Effect-**3**) **248**, the vapor formed in the third evaporation effect (Effect-**3**) **248** is used to drive the evaporation in the fourth evaporation effect (Effect-**4**) **249**, and remainder of the feed seawater **243** in respective evaporation effects is pumped into a subsequent evaporation effect. The vapor formed in the fourth evaporation effect (Effect-**4**) **249** is supplied to the third condenser (C**3**) **228**, where the vapor heats the feed seawater **243**. Upon heating, the vapor is condensed in the third condenser (C**3**) **228** and is

collected in a fresh water tank **250**. Remainder of the feed seawater **243** from the fourth evaporation effect (Effect-**4**) **249** is collected in a brine discharge tank **251**.

(62) In the forward feed configuration, both brine stream and heating vapor stream flow in a same direction from the first evaporation effect (Effect-**1**) **244** to the fourth evaporation effect (Effect-**4**) **249**. Hence, the first evaporation effect (Effect-**1**) **244** operates at a lowest salinity and at a highest temperature. The pressure in the evaporation effects decreases in the flow direction. Each evaporation effect contains heat exchange tubes, vapor space, undiluted feed seawater spray nozzles, mist eliminator, and a last evaporation effect has the brine discharge tank **251**. In each of the second evaporation effect (Effect-**2**) **246** and the third evaporation effect (Effect-**3**) **248**, condensed vapors that run through the heat exchange tubes flow into the fresh water tank **250**.

(63) In the ARC subsystem **203**, the hot molten salt flowing to the generator (G) **220** via the flow path **240** provides thermal energy to separate the weak solution of LiBr—H.sub.2O into the superheated water **252** and the strong solution **253**. The superheated water **252** is cooled in the second condenser (C**2**) **221**. In an aspect, cooling water may be supplied to the second condenser (C**2**) **221** from a cooling tower (not shown). A saturated liquid (obtained after cooling of the superheated water **252**) exiting the second condenser (C**2**) **221** flows through the first throttle valve (TV**1**) **223**, where pressure of the saturated liquid is reduced to a pressure equal to that of the first evaporator (E**1**) **222**. The saturated liquid is evaporated in the first evaporator (E**1**) **222**, thereby providing the first refrigeration effect. Further, saturated water vapor **254** from the first evaporator (E**1**) **222** flows to the absorber (A) **224**.

(64) The strong solution **253** from the generator (G) **220** is pre-cooled in the solution heat exchanger (SHX) **225** and flows to the second throttling valve (TV**2**) **227**, where pressure of the strong solution **253** is reduced to a pressure of the absorber (A) **224**. In an aspect of the present disclosure, the absorber (A) **224** may be maintained at 35° C. and at a pressure of the first evaporator (E**1**) **222**. The saturated water vapor **254** from the first evaporator (E**1**) **222** is mixed with the strong solution **253** at the absorber (A) **224** to form a weak solution (LiBr—H.sub.2O) that is pumped to the solution heat exchanger (SHX) **225** by the fifth pump (P**5**) **226**. The weak solution is pre-heated before flowing into the generator (G) **220**.

(65) Further, power generated by the low-pressure turbine (LPT) **213** is supplied as power input to the high-pressure pump (P**6**) **230** of the RO subsystem **205**. In an aspect, the RO subsystem **205** operates by using the high-pressure pump (P**6**) **230** to increase pressure on a high-salinity side of the RO membrane **231**. Water is pumped by the high-pressure pump (P**6**) **230** through the RO membrane **231**, where the water stream carrying concentrated contaminants (having about 99% of dissolved solids) is collected in the brine discharge tank **251**. Here, for the purpose of brevity, it is assumed that the brine discharge tank **251** of the MED subsystem **204** is same for the RO subsystem **205**, and hence referenced with the same reference numeral. Desalinated water (water stream passing through the RO membrane **231**) is demineralized and is referred to as permeate water **255**.

(66) In an aspect of the present disclosure, superheated steam **256** is supplied from the HRVG **208** to the low-pressure turbine (LPT) **213**, where the superheated steam **256** is expanded. Exhaust steam **257** exiting the low-pressure turbine (LPT) **213** flows to the heat exchanger (HX) **216**, where the exhaust steam **257** is condensed. The condensate is pumped by the third pump (P**3**) **217** from the heat exchanger (HX) **216** to the HRVG **208**, where the condensate is subjected to heating by the heated molten salt flowing through the HRVG **208**.

(67) In the ORC subsystem **202**, condensed water from the first condenser (C**1**) **215** is pumped fourth pump (P**4**) **218** to the heat exchanger (HX) **216**, where the condensed water is heated by the exhaust steam **257** to form steam **258** that is supplied to the turbine (T) **219**. The steam **258** is expanded in the turbine (T) **219** and recondensed in the first condenser (C**1**) **215**. Energy generated by the turbine (T) **219** is used to drive electrochemical reaction of the electrolysis process in the PEM electrolyzer **232**, where a water separation process is carried out to produce hydrogen and

oxygen. In an aspect of the present disclosure, freshwater **259** from the RO subsystem **205** may be used for production of the hydrogen. For example, the freshwater **259** (branching from the permeate water **255**) flows to the first condenser (C**1**) **215**, where the freshwater **259** is heated to a temperature of the PEM electrolyzer **232** by the exhaust from the turbine (T) **219**. The heated freshwater **259** enters the PEM electrolyzer **232** through a flow path **260**.

(68) The produced hydrogen is compressed in a four-stage compressor with an intercooler, such as the hydrogen compressor **233**, to make it suitable for storage. Energy generated by the turbine (T) **219** is also used to drive the hydrogen compressor **233**. In the electrolysis process, hydrogen ions leave a cathode of the PEM electrolyzer **232**, thereby dissipating heat. Hence, the produced hydrogen is cooled by the intercooler to an ambient temperature. The hydrogen is then stored in the hydrogen storage tank **234**. In an aspect of the present disclosure, the hydrogen may be stored at 507 kPa and 85° C. in the hydrogen storage tank **234**. The stored hydrogen is used to produce electric power required to address the additional cooling and freshwater demand. The stored hydrogen is used in PEM fuel cells **261** to produce the additional power. The hydrogen production rate may be modulated according to the available solar energy resulting in a highly efficient, cost-effective, relatively high productivity with a least impact on the environment. The oxygen produced at an anode of the PEM electrolyzer **232** may be stored in the oxygen storage tank **235** for medical use or oxy-fuel combustion requirements.

(69) Further, the MED subsystem **204** may be configured to operate in multiple configurations, such as the forward feed configuration as described with respect to FIG. **2**A-C, a parallel feed configuration, a parallel/cross flow feed configuration, and a backward feed configuration.

(70) FIG. **3**A-C illustrates the parallel feed configuration of the MED subsystem **204** of the system **200**, according to an embodiment of the present disclosure. Except for an arrangement in the MED subsystem **204**, description of the operation of other subsystems remains the same in FIG. **3**A-C, and hence same reference numerals have been used. In the parallel feed configuration, the feed seawater **243** is preheated in the third condenser (C**3**) **228** and is equally distributed in a perpendicular direction to the first evaporation effect (Effect-**1**) **244**, the second evaporation effect (Effect-**2**) **246**, the third evaporation effect (Effect-**3**) **248**, and the fourth evaporation effect (Effect-**4**) **249**. The vapor formed in the first evaporation effect (Effect-**1**) **244** is passed to the second evaporation effect (Effect-**2**) **246** to evaporate the feed seawater **243**, the vapor formed in the second evaporation effect (Effect-**2**) **246** is passed to the third evaporation effect (Effect-**3**) **248** to evaporate the feed seawater **243**, and the vapor formed in the third evaporation effect (Effect-**3**) **248** is passed to the fourth evaporation effect (Effect-**4**) **249** to evaporate the feed seawater **243**. Non-evaporated amount of the feed seawater **243** from each evaporation effect flows to the brine discharge tank **251** and condensed vapor from each evaporation effect (except the first evaporation effect (Effect-**1**) **244**) flows to the fresh water tank **250**.

(71) FIG. **4**A-C illustrates the parallel/cross flow feed configuration of the MED subsystem **204** of the system **200**, according to an embodiment of the present disclosure. Except for an arrangement in the M ED subsystem **204**, description of the operation of other subsystems remains the same in FIG. **4**A-C, and hence same reference numerals have been used. In the parallel/cross flow feed configuration, the feed seawater **243** is preheated in the third condenser (C**3**) **228** and is equally distributed in a perpendicular direction to the first evaporation effect (Effect-**1**) **244**, the second evaporation effect (Effect-**2**) **246**, the third evaporation effect (Effect-**3**) **248**, and the fourth evaporation effect (Effect-**4**) **249**. The vapor formed in the first evaporation effect (Effect-**1**) **244** is passed to the second evaporation effect (Effect-**2**) **246** to evaporate the feed seawater **243**, the vapor formed in the second evaporation effect (Effect-**2**) **246** is passed to the third evaporation effect (Effect-**3**) **248** to evaporate the feed seawater **243**, and the vapor formed in the third evaporation effect (Effect-**3**) **248** is passed to the fourth evaporation effect (Effect-**4**) **249** to evaporate the feed seawater **243**. Brine collected at a bottom of each evaporation effect is pumped into a subsequent evaporation effect. At each evaporation effect, incoming brine mixes with the

non-evaporated amount of the feed seawater **243** and finally the brine is collected in the brine discharge tank **251** from the fourth evaporation effect (Effect-**4**) **249**. Condensed vapor from each evaporation effect (except the first evaporation effect (Effect-**1**) **244**) flows to the fresh water tank **250**.

(72) FIG. **5**A-C illustrates an exemplary instrumentation diagram of the system **200**, according to an embodiment of the present disclosure. Temperature sensors, pressure sensors, and flow sensors are installed in piping of each of the SRC subsystem **201**, the ORC subsystem **202**, the ARC subsystem **203**, the MED subsystem **204**, the RO subsystem **205**, and the hydrogen production subsystem **206** as shown in FIG. **4**A-C. The instrumentations measure the temperature, pressure, flow and transmit the measured values to control and monitoring systems **602** (see FIG. **6**A-C). The control and monitoring systems **602** analyzes and compares the measured value with a predefined setpoint. If the measured value is higher than the predefined setpoint, the control and monitoring systems **602** modulates flow of the working fluid through a control valve until the measured value reaches the predefined setpoint and meets the refrigeration effect, heating effect, freshwater product, and electricity based on the demand.

(73) In an aspect of the present disclosure, a toroidal conductivity sensor may be used in the ARC subsystem **203** to measure the concentration of lithium bromide (LiBr), a mixture of water and lithium bromide (LiBr—H.sub.2O). Water quality sensors are used at intake of the feed seawater **243** and freshwater product for the MED subsystem **204** and the RO subsystem **205** to monitor multiple parameters, such as pH, dissolved oxygen (DO), conductivity, salinity, total dissolved solids (TDS), seawater specific gravity, temperature, turbidity, water depth, and oxidation-reduction potential (ORP). A pH integrated sensor in the water quality sensor is used to measure the amount of alkalinity and acidity in the water. A salinity integrated sensor in the water quality sensor measures the total dissolved salt content in feed seawater **243** or the freshwater products. A TDS integrated sensor in the water quality sensor detects the total dissolved solids in the water and indicates the water quality.

(74) FIG. **6**A-C illustrates a schematic diagram of an advanced control system **600**, according to an aspect of the present disclosure. The advanced control system **600** includes, but not limited to, a touch panel human machine interface **601**, control and monitoring systems **602** in communication with the touch panel human machine interface **601**, terminal blocks **603**, relays **604**, automation controller **605**, timers **606**, input/output (**10**) modules **607**, power outlet **608**, and thermostat and circuit breakers **609**. Additionally, the advanced control system **600** may include advanced controllers, selector switches, pushbutton switches, indication lamps, relays, control transformers, mini circuit breaker, flow meters, pressure differential transmitters, temperature sensor, humidity sensor, wind speed sensor, irradiance sensor, water quality sensor, conductivity sensor, control valves, motorized valve, and interlocking relays. In an aspect of the present disclosure, control cables may be used to connect the sensors of FIG. **4**A-C to the control and monitoring system **602**.

(75) In an aspect of the present disclosure, power consumption by the systems of the present disclosure may be measured with suitable power meter instruments. Measuring the power is one of the most important parameters to check the system's performance, and the power consumption is determined through measurement of the consumed electrical current and is then conveyed to the control and monitoring systems **602**. In an aspect of the present disclosure, advanced control system **600** may include an energy meter used to record each refrigeration effect of each refrigeration system and a power meter used to measure the power consumption of the ARC subsystem **203**. The control and monitoring systems **602** provides data communication for the solar tower system, ensuring reliable system monitoring and straightforward integration into other systems. The control and monitoring systems **602** may include a smart meter, which optimizes self-consumption, records the load curve, and controls the various energy flows with highly accurate measurements and rapid communication via a Modbus interface. Dynamic feed-in control limits may be imposed faster and more accurately than a conventional meter.

(76) In an aspect of the present disclosure, a metrological station may be used and May include multiple sensors for measuring irradiation, ambient temperature, module temperature, wind speed, and amount of thermal energy captured. The metrological station may be integrated into the control and monitoring systems **602** with an internet connection to view in any place.
(77) TABLE-US-00003 TABLE 3 Working fluids Temper- ature Cycle Working Fluid Range (° C.) Power Cycles: 1. Steam Rankine cycle Molten Salt (a mixture of 290-565 With Solar tower sodium nitrate & potassium (SRC) nitrate) 2. Steam Rankine cycle Water (H.sub.2O) 250-400 with HRVG 3. Organic Rankine Cycle Water (H.sub.2O)   50-150 (ORC) Refrigeration Cycle: 1. Absorption Refrigeration Lithium bromide is 10-20 Cycle (ARC) absorbent, and water is a refrigerant (LiBr—H.sub.2O) Desalination Cycles: 1. Multi-Effect Desalination Sea Water. 25-30 (MED) Reverse Osmosis (RO) Sea Water 25-30 Hydrogen Production Cycle 1. Hydrogen (H.sub.2) 40-85 2. Oxygen (O.sub.2) 40 3. Water (H.sub.2O) 30
(78) To this end, the system **200** represents an integration of refrigeration, heating, desalination, hydrogen production, and power generation systems driven by one source of energy (solar energy) to exploit solar energy effectively and to produce a cooling effect and desalinated water in different magnitude for residential application air conditioning, refrigeration, and meet water requirement with the highest efficiency and least environmental impact. The system **200** system also provides a refrigeration effect, a heating effect, the production of freshwater, hydrogen, oxygen, and electricity through integration of different subsystems powered by a renewable energy source, high performance, high refrigeration effect, high water quality, low environmental impact, and relatively low operational costs. Coupling of solar energy technologies with different refrigeration cycles and desalination and hydrogen production provides thermal and electrical energy for complete utilization of the solar energy needed to produce cooling, high-quality freshwater, and electricity.
(79) The system **200** provides a significant solution to produce hydrogen with no adverse environmental impact by integrating hydrogen production system with solar energy and desalination cycles, which is needed to provide the required electrical power and water for the electrochemical reaction of the electrolysis process in the PEM electrolyzer **232**. The system **200** also modulates the use of thermal energy and electrical power according to cooling and freshwater demand. If additional electric power is needed, the stored hydrogen is used in PEM fuel cells **261** to produce the required electricity.
(80) FIG. **7**A-C illustrates an exemplary schematic diagram of a system **700**, according to an aspect of the present disclosure. The system **700** includes a solar-powered steam Rankine cycle (SRC) subsystem **701** (hereinafter referred to as “the SRC subsystem **701**”), an absorption refrigeration cycle (ARC) subsystem **702** (hereinafter referred to as “the ARC subsystem **702**”), a multi-effect desalination (MED) subsystem **703** (hereinafter referred to as “the MED subsystem **703**”), a transcritical refrigeration cycle (TRC) subsystem **704** (hereinafter referred to as “the TRC subsystem **704**”), and a freezing desalination (FRD) subsystem **705** (hereinafter referred to as “the FRD subsystem **705**”). The SRC subsystem **701** is configured to receive solar energy, convert the solar energy into thermal energy, and store the thermal energy. The ARC subsystem **702** is configured to provide a first refrigeration effect with a first range of temperature based on the thermal energy. The MED subsystem **703** is configured to desalinate seawater to produce first desalinated water based on the thermal energy and generate an electrical power based on the thermal energy. The TRC subsystem **704** is configured to generate and store when there is no cooling demand and provide a second refrigeration effect with a second range of temperature based on the electrical energy generated by the MED subsystem **703** and the ice being melted. The FRD subsystem **705** is configured to desalinate the seawater to produce second desalinated water based the ice in the TRC subsystem **704** being melted.
(81) In an aspect of the present disclosure, the SRC subsystem **701** includes a central receiver (CR) **706**, a heat recovery vapor generator (HRVG) **707**, a hot molten salt storage tank **708**, a cold molten salt storage tank **709**, and a heat transfer fluid (HTF) pump (P**1**) **710**. The CR **706** is

configured to receive the solar energy to heat cold molten salt flowing therethrough. The HRVG **707** is configured to generate superheated steam for the MED subsystem **703** based on the heated molten salt and pass the heated molten salt to the ARC subsystem **702**. The hot molten salt storage tank **708** is configured to store the heated molten salt and pass the heated molten salt from the CR **706** to the HRVG **707**. The cold molten salt storage tank **709** is configured to store the cold molten salt and pass the cold molten salt from the ARC subsystem **702** to the CR **706**. The heat transfer fluid (HTF) pump (P**1**) **710** is configured to pump the cold molten salt from the ARC subsystem **702** to the cold molten salt storage tank **709**.

(82) In an aspect of the present disclosure, the ARC subsystem **702** includes a generator (G) **712**, a first condenser (C**1**) **713**, a first evaporator (E**1**) **714**, a first throttle valve (TV**1**) **715**, an absorber (A) **716**, a solution heat exchanger (SHX) **717**, a third pump (P**3**) **718**, and a third throttling valve (TV**3**) **719**. The generator (G) **712** is configured to pass the heated molten salt from the HRVG **707** to the cold molten salt storage tank **709** of the SRC subsystem **701**; generate superheated water vapor refrigerant based on the heated molten salt flowing through the generator (G) **712**; and generate a strong solution based on a weak solution flowing into the generator (G) **712**. The first condenser (C**1**) **713** is configured to generate saturated liquid refrigerant by cooling the superheated water vapor refrigerant from the generator (G) **712**. The first evaporator (E**1**) **714** is configured to provide the first refrigeration effect with the first range of temperature by generating saturated water vapor based on the saturated liquid refrigerant from the first condenser (C**1**) **713**. The first throttle valve (TV**1**) **715** is configured to pass the saturate liquid refrigerant from the first condenser (C**1**) **713** to the first evaporator (E**1**) **714** and reduce a pressure of the saturated liquid refrigerant to a pressure of the first evaporator (E**1**) **714**. The absorber (A) **716** is configured to generate the weak solution by mixing the strong solution with the saturated water vapor from the first evaporator (E**1**) **714**. The solution heat exchanger (SHX) **717** is configured to pre-heat the weak solution from the absorber (A) **716**, pass the pre-heated weak solution to the generator (G) **712**, and pre-cool the strong solution from the generator (G) **712**. The third pump (P**3**) **718** is configured to pump the weak solution from the absorber (A) **716** to the solution heat exchanger (SHX) **717**. The third throttling valve (TV**3**) **719** is configured to pass the pre-cooled strong solution from the solution heat exchanger (SHX) **717** to the absorber (A) **716** and reduce a pressure of the pre-cooled strong solution to a pressure of the absorber (A) **716**.

(83) In an aspect of the present disclosure, the MED subsystem **703** includes a steam turbine (T) **720**, a steam ejector (EJE) **721**, a second condenser (C**2**) **722**, and a second pump (P**2**) **723**. The steam turbine (T) **720** is configured to generate the electrical power based on the superheated steam from the HRVG **707** of the SRC subsystem **701**. The steam ejector (EJE) **721** is configured to generate a mixture by mixing an exhaust from the steam turbine (T) **720** and vapor from the MED subsystem **703** and pass the mixture to the MED subsystem **703**. The second condenser (C**2**) **722** is configured to heat and forward seawater to a first evaporation effect of a plurality of evaporation effects. The plurality of evaporation effects is configured to produce fresh water from the heated seawater, where the first evaporation effect is driven by the mixture from the steam ejector (EJE) **721**. The second pump (P**2**) **723** is configured to pump condensate from the first evaporation effect to the HRVG **707** of the SRC subsystem **701**.

(84) In an aspect of the present disclosure, the TRC subsystem **704** includes a second evaporator (E**2**) **724**, an internal heat exchanger (IHX) **725**, a compressor (COMP) **726**, a third condenser (C**3**) **727**, a second throttling valve (TV**2**) **728**, and an ice storage system **729**. The second evaporator (E**2**) **724** is configured to provide the second refrigeration effect with the second range of temperature by generating refrigerant vapor. The internal heat exchanger (IHX) **725** is configured to generate superheated refrigerant vapor by evaporating liquid droplets of the refrigerant vapor from the second evaporator (E**2**) **724**. The compressor (COMP) **726** is powered by the electrical power generated from the steam turbine (T) **720** of the MED subsystem **703** and is configured to generate high-pressure superheated refrigerant vapor by compressing the superheated refrigerant

vapor from the internal heat exchanger (IHX) **725**. The third condenser (C**3**) **727** is configured to cool the high-pressure superheated refrigerant vapor to saturated liquid refrigerant by melting ice in the FRD subsystem **705**. The second throttling valve (TV**2**) **728** is configured to pass the saturated liquid refrigerant to the second evaporator (E**2**) **724** and reduce a pressure of the saturated liquid refrigerant to a pressure of the second evaporator (E**2**) **724**. The ice storage system **729** is configured to build the ice when there is no cooling demand and melt the ice when there is cooling demand.

(85) In an aspect of the present disclosure, the FR D subsystem **705** includes a washer **730** configured to separate ice and brine that are transported from the ice storage tank **729** of the TRC subsystem **704**; and a fourth condenser (C**4**) **731** configured to pre-cool seawater through a product fresh water steam from an ice melter and discharged brine steam from the washer **730**.

(86) In operation of the system **700**, solar radiation incident on a heliostat field **732** is reflected to the CR **706** that is fixed on a solar tower **733**. Thermal energy from the reflected solar radiation is used to heat the cold molten salt contained in the CR **706**. In an aspect of the present disclosure, the CR **706** may be coated with a selective coating to minimize reflectivity, thereby improving absorption performance. The heated molten salt flows from the CR **706** to the hot molten salt storage tank **708** via a flow path **734**. The heated molten salt may be stored by the hot molten salt storage tank **708** and may be subsequently passed to the HRVG **707** via a flow path **735**. Heat transfer between the heated molten salt and the water occurs at the HRVG **707**, where superheated steam **737** is generated for the MED subsystem **703**. The generated superheated steam is used to drive the steam turbine **720**. The molten salt, after the heat transfer at the HRVG **707**, flows to the generator (G) **712** via a flow path **736**. The molten salt then flows to the cold molten salt storage tank **709**. Solar energy is stored in two tanks to aid continuous operation of the system **700** during unavailability of the solar energy.

(87) The superheated steam **737** is expanded in the steam turbine (T) **720** and exhaust steam **738** from the steam turbine (T) **720** flows to the steam ejector (EJE) **721**, thereby creating high vacuum pressure in the steam ejector (EJE) **721**. Such high vacuum pressure draws vapor from the MED subsystem **703**. The steam ejector (EJE) **721** generates the mixture by mixing the exhaust steam **738** from the steam turbine (T) **720** and vapor from the MED subsystem **703**. The mixture then flows to the MED subsystem **703**. FIG. **7** also illustrates a forward feed configuration of the MED subsystem **703**, among other configurations.

(88) Feed seawater **739** enters the second condenser (C**2**) **722** and is preheated by vapor flowing through the second condenser (C**2**) **722**. The preheated feed seawater **739** flows to the first evaporation effect (Effect-**1**) **740** of a plurality of evaporation effects, where the preheated feed seawater **739** is sprinkled therein. The mixture from the steam ejector (EJE) **721** is associated with sufficient heat to evaporate the sprinkled feed seawater **739** in the first evaporation effect (Effect-**1**) **740**. Vapor from the first evaporation effect (Effect-**1**) **740** is supplied to the second evaporation effect (Effect-**2**) **741**. Non-evaporated feed seawater **739** collects at a bottom of the first evaporation effect (Effect-**1**) **740** and is pumped into the second evaporation effect (Effect-**2**) **741**, where the non-evaporated feed seawater **739** is sprinkled. The vapor is routed through tubes in the second evaporation effect (Effect-**2**) **741** and drives the evaporation of the sprinkled feed seawater **739** in the second evaporation effect (Effect-**2**) **741**. The mixture in the first evaporation effect (Effect-**1**) **740** is condensed and pumped from the first evaporation effect (Effect-**1**) **740** to the HRVG **707**, by the second pump (P**2**) **723**. Condensed mixture (water) is heated by the hot molten salt in the HRVG **707** to form the superheated steam **737**. Further, vapor formed in the second evaporation effect (Effect-**2**) **741** is passed into the third evaporation effect (Effect-**3**) **742** and vapor formed in the third evaporation effect (Effect-**3**) **742** is passed into the fourth evaporation effect (Effect-**4**) **743**. In an aspect of the present disclosure, each of the first evaporation effect (Effect-**1**) **740**, the second evaporation effect (Effect-**2**) **741**, the third evaporation effect (Effect-**3**) **742**, and the fourth evaporation effect (Effect-**4**) **743** may include an array of sprinkler nozzles to

sprinkle the feed seawater **739**.

(89) The steam ejector (EJE) **721** operates based on the exhaust steam **738** and entrains a portion of vapor formed in the second evaporation effect (Effect-**2**) **741**. The steam ejector (EJE) **721** is configured to compress the entrained vapor from the second evaporation effect (Effect-**2**) **741** by the exhaust steam **738** to form the mixture associated with a desired temperature.

(90) Vapor formed in the fourth evaporation effect (Effect-**4**) **743** flows to the second condenser (C**2**) to preheat the incoming feed seawater **739**. After losing heat to the feed seawater **739**, the vapor is condensed in the second condenser (C**2**) to form freshwater **759**. Additionally, the vapor condensed in each of the second evaporation effect (Effect-**2**) **741**, the third evaporation effect (Effect-**3**) **742**, and the fourth evaporation effect (Effect-**4**) **743** is collected as the freshwater **759**. Non-evaporated feed seawater **739** (also referred to as the brine) from the fourth evaporation effect (Effect-**4**) **743** is routed to a brine discharge **760**.

(91) In the forward feed configuration, both brine and heating vapor streams flow in the same direction from the first evaporation effect (Effect-**1**) **740** to the fourth evaporation effect (Effect-**4**) **743**, where pressure and temperature are reduced. The first evaporation effect (Effect-**1**) **740** operates at a lowest salinity and a highest temperature and pressure. The undiluted feed seawater **739** leaving an evaporation effect is sprayed in a subsequent effect. The pressure in the evaporation effects decreases in a flow direction. In an aspect of the present disclosure, each evaporation effect contains heat exchange tubes, vapor space, undiluted feed seawater spray nozzles, mist eliminator.

(92) In the ARC subsystem **702**, the molten salt flowing along the flow path **736** enters the generator (G) **712**, where the molten salt provides the thermal energy required to separate the weak solution of LiBr—H.sub.2O into superheated water vapor refrigerant **744** and a strong solution **745**. The superheated water **744** is cooled in the first condenser (C**1**) **713** to form saturated liquid refrigerant **746**. In an aspect of the present disclosure, cooling water from a cooling tower (not shown) may be supplied to the first condenser (C**1**) **713** to cool the superheated water vapor refrigerant **744**. The saturated liquid refrigerant **746** flows through the first throttle valve (TV**1**) **715**, where a pressure of the saturated liquid refrigerant **746** is reduced to a value equal to a pressure of the first evaporator (E**1**) **714**. In an aspect of the present disclosure, the first evaporator (E**1**) **714** is configured to provide the first refrigeration effect with the first range of temperature by generating saturated water vapor **747** based on the saturated liquid refrigerant **746** from the first condenser (C**1**) **713**.

(93) The saturated water vapor **747** flows into the absorber (A) **716**. The strong solution **745** from the generator (G) **712** is pre-cooled in the solution heat exchanger (SHX) **717** and further flows through the third throttling valve (TV**3**) **719**, where a pressure value of the pre-cooled strong solution **745** is reduced to a pressure of the absorber (A) **716**. In an aspect of the present disclosure, the absorber (A) **716** may be maintained at a temperature of about 35° C. The saturated water vapor **747** flowing into the absorber (A) **716** is mixed with the pre-cooled strong solution **745** to form the weak solution **748** (such as LiBr—H.sub.2O) that is pumped back to the solution heat exchanger (SHX) **717** by the third pump (P**3**) **718**. The weak solution **748** is preheated in the solution heat exchanger (SHX) **717** prior to being supplied to the generator (G) **712**.

(94) The energy produced by the steam turbine (T) **720** of the MED subsystem **703** is supplied as power input to the TRC subsystem **704** having the ice storage system **729** coupled to the FRD subsystem **705**. In the TRC subsystem **704**, a water-based solution is cooled during the nighttime. The N.sub.2O refrigerant provides a very low-temperature range and refrigeration effect that can be used for deep freezing applications in the range of −50° C. to −80° C. In an aspect, N.sub.2O refrigerant vapor **749** flows to the internal heat exchanger (IHX) **725**, where liquid portion is evaporated to form superheated refrigerant vapor **750** that is compressed to high refrigerant pressure and temperature in the compressor (COMP) **726**. A high-pressure superheated refrigerant vapor **751** from the compressor (COMP) **726** flows to the third condenser (C**3**) **727** and is cooled to form liquid refrigerant **753**. The hot vapor from the third condenser (C**3**) **727** is routed to the ice

melter **752** of the FRD subsystem **705**, where the hot vapor melts the ice therein. Utilization of the heat in the third condenser (C**3**) **727** is improves an overall efficiency of the system **700** due to reducing the required power needed to provide cooling to the third condenser (C**3**) **727** and provides heating to the ice melter **752** to melt the ice and produce fresh water.

(95) The liquid refrigerant **753** from the third condenser (C**3**) **727** flows to the internal heat exchanger (IHX) **725** for additional cooling and then flows through the second throttling valve (TV**2**), where a pressure of the liquid refrigerant **753** is reduced to a value equal to a pressure of the second evaporator (E**2**) **724**. The internal heat exchanger (IHX) **725** enhances coefficient of performance of the TRC subsystem **704** owing to the increase in temperature of the superheated refrigerant vapor **750** entering the compressor (COMP) **726**, which improves superheating and decreases temperature of the liquid refrigerant **753** flowing to the second evaporator (E**2**) **724**, thereby improving subcooling. The second evaporator (E**2**) **724** achieves the second refrigeration effect with the second range of temperature measured as a difference between the temperature of the liquid refrigerant **753** and the N.sub.2O refrigerant vapor **749**.

(96) The TRC subsystem **704** generates ice during no cooling demand. The stored thermal energy of the ice storage system **729** is used to meet changing cooling load demands. In peak-time and grid outages, the stored energy in the ice storage tank **729** works continuously to cool the building adding an increased level of redundancy.

(97) Ice in the ice storage system **729** melts to meet the cooling demand, and a secondary glycol pump (SP) **754** is actuated. Melting cycle begins as hot water **755** enters the system **700** from the building. The hot water **755** passes through a heat exchanger (HX) **756** to lose heat and form cold chilled water **757** that circulates through a tertiary glycol pump (TP) **758** to meet the cooling demand from the building. It will be apparent from FIG. **7**A-C that there is no direct contact between the melting ice and hot water **755**, which is cooled indirectly in the heat exchanger (HX) **756** by melting the ice at the ice storage system **729**.

(98) The FRD subsystem **705** is an indirect type, and saline water does not contact the refrigerant or chilled water. The feed seawater **739** flows to the fourth condenser (C**4**) **731** and is pre-cooled to desired temperature with aid of product freshwater stream **759** from the ice melter **752** and a discharged brine stream **760** from the washer **730**. The pre-cooled feed seawater **739** is routed to the ice storage system **729** via a flow path **762**. Ice crystal is formed on an outer surface of the ice storage system **729**, and MV-**3** opens during the desalination process. In an aspect of the present disclosure, chilled water flows inside a tube of a cooling freezing chamber **763** to freeze the pre-cooled feed seawater **739** around the tube. The ice and brine are transported to the washer **730** via a flow path **764**. The washer **730** is configured to separate the ice and the brine. The ice is transported to the ice melter **752**, where the ice is melted by the heat of condensation released from hot water/vapor supplied from the third condenser (C**3**) **727**. A small amount of the freshwater **759** from the ice melter **752** is bypassed to the washer **730** to wash the ice crystals, and a majority of the freshwater **759** is routed to the fourth condenser (C**4**) **731** to cool the feed seawater **739**. The freshwater **759** is then collected in a freshwater storage tank **765**. Similarly, the brine discharged from the washer **730** is collected in the brine tank **766** after passing through the fourth condenser (C**4**) **731**.

(99) In an aspect of the present disclosure, ethylene/propylene glycol may be used in the chilled water routed through the cooling freezing chamber **763** for freezing protection. Advanced-grade glycol is designed to prevent corrosion in refrigeration systems. Inhibitors may be added to the chilled water to prevent the ethylene glycol from becoming acidic and to provide additional protection for metal components in the thermal storage system.

(100) TABLE-US-00004 TABLE 4 Sequence of Operation of Transcritical Refrigeration, Freezing Desalination, and Ice Storage System Operation Mode TRC P7 P8 MV-1 MV-2 MV-3 CV-1 FRD only On On Off X1-X2 Y1-Y3 Z1-Z3 Closed FRD and On On On X1-X2 Variable Z1-Z3 Closed Cooling FRD, Cooling On On On X1-X2 Variable Z1-Z2 Open with Ice Build Cooling-Ice On On

On Variable Y1-Y2 Z1-Z2 Closed with TRC Cooling-TRC On On On X1-X3 Y1-Y2 Closed Closed Only Cooling-Ice Off On On Variable Y1-Y2 Z1-Z2 Closed Only Ice Build On On Off X1-X2 Y1-Y3 Z1-Z2 Closed (Cooling Chamber) Ice Build On On On X1-X2 Variable Z1-Z2 Closed with Cooling

(101) Table-41 illustrates operation modes of the TRC subsystem **704** with the ice storage system **729**, and the FRD subsystem **705**. The sequence of operation and modulation of motorized valves (MV-**1**, MV-**2** MV-**3**, and CV-**1**) depends on cooling and freshwater demand. As listed in the table, the TRC subsystem **704** with the ice storage system, and the FRD subsystem **705** may operate in eight different modes.

(102) In a first mode, the integrated system [the TRC subsystem **704** and the ice storage system **729**] with the FR D subsystem **705** provides a low-temperature chilled water-glycol needed for the FRD subsystem **705**. In this mode, the TRC subsystem **704** and P**7** pump is actuated, and the secondary glycol pump (SP) **754** is turned off due to no cooling demand. All chilled water-glycol is used for the FRD subsystem **705**. The chilled water-glycol passes through the MV-**1**, and then the chilled water flows to a desalination freezing chamber **767** of the ice storage system **729** through the MV-**3**, where the CV-**1** remains closed. The warm chilled water leaves the desalination freezing chamber **767** and passes through the MV-**2**. In an aspect of the present disclosure, an advanced low-pressure air distribution system **768** may be provided at the desalination freezing chamber **767** to create air bubble that is used to enhance heat transfer during the ice build cycle. When a thickness of the ice reaches a design value, the advanced control system **1100** (see FIG. **11**) sends a signal to turn off the TRC subsystem **704**. The warm water is pumped to the P**7** pump, and then returns to the second evaporator (E**2**) **724**. Low-temperature glycol-water **769** in the desalination freezing chamber **767** is used to form the ice crystal of the feed seawater **739** on the external surface of the tubes at the desalination freezing chamber **767** of the ice storage system **729**. In an aspect of the present disclosure, the desalination freezing chamber **767** may include a steel perforated base plate configured to separate the desalination freezing chamber **767** into two parts, where a lower part is configured to collect the drained brine and ice wash, and an upper part associated with the glycol cycle to facilitate freezing process therein. In an aspect of the present disclosure, the desalination freezing chamber **767** and the cooling freezing chamber **763** may be designed to include smaller coil diameter to achieve thick ice formation (for example, about 2 inches) and thereby improve efficiency of the TRC subsystem **704**.

(103) In a second mode, the system **700** provides low-temperature chilled water required for freezing desalination and cooling. The TRC subsystem **704**, the P**7** pump, and the secondary glycol pump (SP) **754** is actuated, and the glycol-water flows through the MV-**1**, then the chilled water flows to desalination freezing chamber **767** through the MV-**3**, where the CV-**1** remains closed. The chilled water modulates at the MV-**2** to meet the required cooling demand. The low-temperature glycol-water **769** is passed to the desalination freezing chamber **767**, where low-temperature chilled glycol-water **769** is used to form the ice crystal of the feed seawater **739** on the external surface of the tubes at the desalination freezing chamber **767**. The advanced low-pressure air distribution system **768** is disposed at the desalination freezing chamber **767** to create air bubbles used to enhance heat transfer during the ice build cycle. When a thickness of the ice reaches a design value, the advanced control system **1100** (see FIG. **11**) sends a signal to turn off the TRC subsystem **704**.

(104) In a third mode, the system **700** provides low-temperature chilled water for freezing desalination and cooling with building ice for cooling purposes. The TRC subsystem **704**, the P**7** pump, and the secondary glycol pump (SP) **754** is actuated, and the chilled water flows through the MV-**1**, then flows to the cooling freezing chamber **763** and the desalination freezing chambers **767** through the MV-**3**, and the CV-**1** remains open to pass the chilled water to the desalination freezing chamber **767**. The chilled water is also used in the cooling freezing chamber **763** and the desalination freezing chambers **767** to generate and store for cooling purposes. The advanced low-

pressure distribution system **768** is disposed at the cooling freezing chamber **763** and the desalination freezing chambers **767** to create air bubble that is used to enhance heat transfer during the ice build cycle. When a thickness of the ice reaches a design value, the advanced control system **1100** (see FIG. **11**) sends a signal to turn off the TRC subsystem **704**. The chilled water modulates at the MV-**2** according to the required cooling demand. The chilled water passes through the CV-**1**, where low-temperature glycol-water flows to the desalination freezing chamber **767** to form the ice-crystal of the feed seawater **739** on the external surface of the tubes at the desalination freezing chamber **767**.

(105) In a fourth mode, the system **700** provides low-temperature chilled water needed cooling and builds ice during low-ambient operation and nighttime at the cooling freezing chamber **763** to meet the cooling requirement when cooling load exists. The TRC subsystem **704**, the P**7** pump, and the secondary glycol pump (SP) **754** is actuated, and the chilled water modulates at the MV-**1** to regulate the supplied cooling from the TRC subsystem **704** and the cooling freezing chamber **763**, and then the chilled water flows to cooling freezing chambers **763** through the MV-**3**, and the CV-**1** remains closed. The low temperature glycol-water is passed to the cooling freezing chamber **763** to generate and store and store the ice for cooling purposes. the advanced low-pressure distribution system **768** is disposed at the cooling freezing chamber **763** to create air bubbles used to enhance heat transfer during the ice build cycle. When a thickness of the ice reaches a design value, the advanced control system **1100** (see FIG. **11**) sends a signal to turn off the TRC subsystem **704**. The glycol-water passes **769** through the MV-**2**, and it is pumped to the HX **756**. The low-temperature glycol-water **769** is pumped to the secondary glycol pump (SP) **754** and then flows through the HX **756** to meet the cooling requirements.

(106) In a fifth mode, the system **700** provides cooling by the TRC subsystem **704** only. The TRC subsystem **704**, the P**7** pump, and the secondary glycol pump (SP) **754** remains actuated, the MV-**3** and the CV-**1** remains closed. The glycol-water **769** is bypassed through the MV-**1** to the MV-**2**. Then the chilled water passes through the MV-**2** and is pumped to the HX **756**. The low-temperature glycol-water **769** is pumped to the secondary glycol pump (SP) **754** and then flows through the HX **756** to meet the cooling demand. The advanced control system **1100** controls the operations of the TRC subsystem **704** to meet the exact cooling demands.

(107) In a sixth mode, the system **700** provides cooling by the stored ice at the cooling freezing chamber **763** at the ice storage system **729**. The TRC subsystem **704** is turned off, and the P**7** pump, and the secondary glycol pump (SP) **754** is kept running. The glycol-water is bypassed through the MV-**1**, and then the chilled water flows to the cooling freezing chamber **763** through the MV-**3**, where the CV-**1** remains closed. The chilled water then passes through the MV-**2** and is pumped to the HX **756**. The advanced low-pressure distribution system **768** is disposed at the cooling freezing chamber **763** to create air bubble uses to enhance heat transfer during the ice build cycle. When the cooling demand exists, the low-temperature glycol-water **769** is pumped to the secondary glycol pump (SP) **754** and then flows through the HX **756** to meet the cooling demand while TRC subsystem **704** is turned off. The melt-out cycle begins. Warm water from the HX **756** passes to the cooling freezing chamber **763** and is cooled indirectly with the melting ice. During the ice melting cycle, the melted water is agitated by the advanced low-pressure distribution system **768** to produce a uniform ice melting process and improve heat transfer.

(108) In the seventh mode, the system **700** builds ice only during low-ambient operation conditions and when no cooling load exists. The TRC subsystem **704** and the P**7** pump remains running, and the secondary glycol pump (SP) **754** is turned off. The low-temperature glycol-water **769** passes through the MV-**1**. The chilled water flows to cooling freezing chamber **763** through the MV-**3**, where the CV-**1** remains closed. The chilled water builds ice on the outer surface of the tube and stores it at the cooling freezing chamber **763** to be used when a cooling load exists. the advanced low-pressure distribution system **768** is disposed at the cooling freezing chamber **763** to create air bubble that is used to enhance heat transfer during the ice build cycle. When a thickness of the ice

reaches a design value, the advanced control system **1100** (see FIG. **11**) sends a signal to turn off the TRC subsystem **704**. The warm, glycol-water returns to the second evaporator (E**2**) **724**, where TRC subsystem **704** is feeding the refrigeration effect at the second evaporator (E**2**) **724**. The warm glycol-water passes through the MV-**2** and is pumped back to the second evaporator (E**2**) **724** to be chilled, and the ice build cycle is repeated.
(109) In an eighth mode, the system **700** provides cooling and builds ice at the cooling freezing chamber **763** at the ice storage system **729** to be used for cooling purposes. The TRC subsystem **704**, the P**7** pump, and the secondary glycol pump (SP) **754** remains running, and the glycol-water flows through the MV-**1**, and then the chilled water flows to the cooling freezing chambers **763** through the MV-**3**, and the CV-**1** remains closed. The chilled water is used in the cooling freezing chamber **763** to generate and store. The advanced low-pressure distribution system **768** is disposed at the cooling freezing chamber **763** to create air bubble that is used to enhance heat transfer during the ice build cycle. When a thickness of the ice reaches a design value, the advanced control system **1100** (see FIG. **11**) sends a signal to turn off the TRC subsystem **704**. The chilled water modulates at the MV-**2** according to the required cooling demand.
(110) The MED subsystem **703** is configured to operate in one of a parallel feed configuration, the forward feed configuration as described with reference to FIG. **7**A-C, a backward feed configuration, and a parallel/cross flow feed configuration. FIG. **8**A-C illustrates the parallel feed configuration of the MED subsystem **703**, while FIG. **9**A-C illustrates the parallel/cross flow feed configuration of the MED subsystem **703**, and FIG. **10**A-C illustrates an exemplary instrumentation diagram of FIG. **7**A-C, according to aspects of the present disclosure. These configurations are described with respect to FIG. **3**A-C, FIG. **4**A-C, and FIG. **5**A-C respectively, and hence not repeated here for the purpose of brevity.
(111) FIG. **11** illustrates an exemplary advanced control system **1100**, according to an aspect of the present disclosure. In an aspect of the present disclosure, the advanced control system **1100** includes, but is not limited to, a control and monitoring system **1102**, terminal blocks **1104**, a touch panel human machine interface **1106**, relays **1108**, timers **1110**, thermostats and circuit breakers **1112**, a stepdown transformer **1114**, a power outlet **1116**, I/O modules **1118**, and an automation controller **1120**. In an aspect of the present disclosure, the advanced control system **1100** may further include an advanced controller to enable the sequence of operations and provide a fully functional integrated system. Further, a control system of the advanced control system **1100** may include advanced controllers, selector switches, pushbutton switches, indication lamps, relays, control transformers, mini circuit breaker, flow meters, pressure differential transmitters, temperature sensor, humidity sensor, wind speed sensor, irradiance sensor, water quality sensor, conductivity sensor, control valves, motorized valve, and interlocking relays. In an aspect of the present disclosure, control cables may be used to connect the sensors with the control system.
(112) In an aspect of the present disclosure, the control system may monitor and control the ice building cycle operation in the desalination freezing chamber **767** and the cooling freezing chamber **763**. When the ice build cycle begins either in the desalination freezing chamber **767** and the cooling freezing chamber **763**, the TRC subsystem **704** may operate at a maximum refrigeration capacity without unloading until the cooling freezing chamber **763** is fully charged. When the cooling freezing chamber **763** is fully charged, the control system may send a signal to turn off the TRC subsystem **704**, and the TRC subsystem **704** may not resume operation until cooling or freshwater demands exist. The control system may include advanced features, such as termination of the operation at a low water temperature. The low-temperature cutout prevents the ice build from starting if enough water is not present in both freezing chambers.
(113) TABLE-US-00005 TABLE 5 Working fluids for the system 700 Temper- ature Cycle Working Fluid Range (° C.) Power Cycle: 1. Steam Rankine cycle Molten Salt (a mixture of 290~565 With Solar tower (SRC). sodium nitrate & potassium nitrate). 2. Steam Rankine cycle Water (H.sub.2O). with HRVG. Multi-Effect Refrigeration Cycle 1. Transcritical Nitrous Oxide

($N.sub.2O$). −50-−80 Refrigeration cycle (TRC) 2. Absorption Refrigeration Lithium bromide is 10-20 Cycle (ARC) absorbent, and water is a refrigerant (LiBr—$H.sub.2O$). Desalination Cycles: 1. Multi-Effect Desalination Sea Water. 25-30 (MED). 2. Freezing Desalination Sea Water 25-30 (FRD). Ice Storage System: 1. Desalination Freezing Ethylene/Propylene Glycol −5.56 Chamber. 2. Cooling Freezing Ethylene/Propylene Glycol −6.67 Chamber. 3. HX-Cooling Load Chilled Water 4 to 12

(114) To this end, it will be understood that the system **700** represents an integration of the ARC subsystem **702**, the MED subsystem **703**, the TRC subsystem **704**, and the FRD subsystem **705** which exploits the solar energy effectively and produces a cooling effect in different magnitude and temperature ranges for refrigeration applications, vaccine storage, food preservation and deep freezing with the highest efficiency and least environmental impact. The system **700** produces useful electric power with zero $CO.sub.2$ emissions and with no adverse impact on the environment due to usage of renewable energy sources.

(115) In an aspect of the present disclosure, wind energy, biomass, and hydro energy may be coupled with the system **700** to produce the electricity needed to operate the pumps or the advanced control system **1100**. In an aspect of the present disclosure, a gain output ratio (GOR) of the system **700** may be enhanced by adding more evaporation effects than producing more freshwater. In an aspect of the present disclosure, the MED subsystem **703** may be replaced by a small-scale Multi-Stage Flashing Desalination System (MSF), which operates at 110° C. temperature and may serve medium to large community water needs. In an aspect of the present disclosure, the ARC subsystem **702** may operate as a heat pump to provide a heating effect for a residential and commercial heating system.

(116) As used herein, the terms “a” and “an” and the like carry the meaning of “one or more.”

(117) Obviously, numerous modifications and variations of the present invention are possible in light of the above teachings. It is, therefore, to be understood that, within the scope of the appended claims, the invention may be practiced otherwise than as specifically described herein.

Claims

1. A system, comprising: a solar-powered steam Rankine cycle (SRC) subsystem configured to receive solar energy, convert the solar energy into thermal energy, and store the thermal energy; an ejector refrigeration cycle (ERC) subsystem configured to generate an electrical power based on the thermal energy, and provide a first refrigeration effect with a first range of temperature based on the thermal energy; an absorption refrigeration cycle (ARC) subsystem configured to provide a second refrigeration effect with a second range of temperature based on the thermal energy; a brine refrigeration cycle (BRC) subsystem configured to generate and store when there is no cooling demand, and provide a third refrigeration effect with a third range of temperature based on the electrical power generated by the ERC subsystem and the ice being melted; and an adsorption refrigeration cycle (ADRC) subsystem configured to provide a fourth refrigeration effect with a fourth range of temperature based on the thermal energy, wherein the SRC comprises a heat recovery vapor generator (HRVG) configured to generate superheated steam for the ERC subsystem; wherein the ERC subsystem includes a steam turbine (T) configured to generate the electrical power based on the superheated steam from the HRVG of the SRC subsystem; and wherein the BRC subsystem includes a compressor (COMP) powered by the electrical power generated from the steam turbine of the ERC subsystem and configured to generate high-pressure superheated refrigerant vapor.

2. The system of claim 1, wherein the SRC subsystem includes a central receiver (CR) configured to receive the solar energy to heat cold molten salt flowing through the central receiver; wherein the heat recovery vapor generator (HRVG) is configured to generate the superheated steam based on the heated molten salt, and pass the heated molten salt to the ARC subsystem; a hot molten salt

storage tank configured to store the heated molten salt, and pass the heated molten salt from the CR to the HRVG; a cold molten salt storage tank configured to store the cold molten salt, and pass the cold molten salt from the ADRC subsystem to the CR; and a heat transfer fluid (HTF) pump configured to pump the cold molten salt from the cold molten salt storage tank to the CR.
3. The system of claim 2, wherein the ERC subsystem includes a first evaporator (E**1**) configured to provide the first refrigerant effect with the first range of temperature by evaporating water flowing through the first evaporator into water vapor; a steam ejector configured to generate a mixture by mixing an exhaust from the steam turbine and the water vapor from the first evaporator; a first condenser (C**1**) configured to cool the mixture from the steam ejector into the water; a first throttling valve (TV**1**) configured to pass a first volume of the water from the first condenser to the first evaporator, and reduce a pressure of the first volume of the water to a pressure of the first evaporator; and a first pump (P**1**) configured to pump a second volume of the water from the first condenser to the HRVG of the SRC subsystem.
4. The system of claim 3, wherein the BRC subsystem includes a third evaporator (E**3**) configured to provide the third refrigeration effect with the third range of temperature by evaporating liquid refrigerant flowing through the third evaporator into superheated refrigerant vapor; wherein the compressor is configured to generate high-pressure superheated refrigerant vapor by compressing the superheated refrigerant vapor from the third evaporator; a third condenser (C**3**) configured to cool the high-pressure superheated refrigerant vapor into the liquid refrigerant flowing to the third evaporator; a fourth throttling valve (TV**4**) configured to pass the liquid refrigerant from the third condenser to the third evaporator, and reduce a pressure of the liquid refrigerant to a pressure of the third evaporator; and an ice storage system configured to build the ice when there is no cooling demand and melt the ice when there is cooling demand.
5. The system of claim 2, wherein the ARC subsystem includes a generator (G) configured to pass the heated molten salt from the HRVG of the SRC subsystem to adsorption chambers of the ADRC subsystem, generate superheated water vapor refrigerant based on the heated molten salt flowing through the generator, and generate a strong solution based on a weak solution flowing into the generator; a second condenser (C**2**) configured to generate saturated liquid refrigerant by cooling the superheated water vapor refrigerant from the generator; a second evaporator (E**2**) configured to provide the second refrigeration effect with the second range of temperature by generating saturated water vapor based on the saturated liquid refrigerant from the second condenser; a second throttle valve (TV**2**) configured to pass the saturated liquid refrigerant from the second condenser to the second evaporator, and reduce a pressure of the saturated liquid refrigerant to a pressure of the second evaporator; an absorber (A) configured to generate the weak solution by mixing the strong solution with the saturated water vapor from the second evaporator; a solution heat exchanger (SHX) configured to pre-heat the weak solution from the absorber, pass the pre-heated weak solution to the generator, and pre-cool the strong solution from the generator; a second pump (P**2**) configured to pump the weak solution from the absorber to the SHX; and a third throttling valve (TV**3**) configured to pass the pre-cooled strong solution from the SHX to the absorber, and reduce a pressure of the pre-cooled strong solution to a pressure of the absorber.
6. The system of claim 5, wherein the ADRC subsystem includes two adsorption chambers (AC**-1** and AC-**2**) configured to alternatively and respectively perform as adsorber and desorber; a fourth evaporator (E**4**) configured to provide the fourth refrigerant effect with the fourth range of temperature by evaporating water flowing through the four evaporator into water vapor flowing to a first adsorption chamber of the two adsorption chambers performing as the absorber; and a fourth condenser (C**4**) configured to cool the water vapor from a second adsorption chamber of the two adsorption chambers performing as the desorber, wherein the heated molten salt is cooled through the two adsorption chambers and returns to the cold molten salt storage tank of the SRC subsystem.